Norm-fiber densities and relative traces in characteristic two

Wencong Huang Baohua Li

November 2, 2026

Abstract

We study linked multiplicative orders in quartic norm-one subgroups in characteristic two. Put $t = 2^s$, $s \geq 2$, and let $C_d \leq \mathbb{F}_{t^4}^\times$ have order $d = (t+1)(t^2+1)$. Genus-one norm fibers give effective joint-generation and double full-order densities in this growing-index family. The respective leading terms are $(t+1)^2 \prod_{p|d}(1-p^{-2})$ and $(\varphi(d)/d)^2(t+1)^2$. For every parameter, a double full-order pair with prescribed first inner norm b exists exactly when b has order $t+1$. A three-coordinate prime sieve, a uniform tail bound and complete finite certificates prove this support statement, including composite t^2+1. Across primitive inner norms, an exact Kloosterman formula gives unsieved variance asymptotic to t. For odd s, the cubic coordinate of the second-norm sieve is a genus-two trace; a projective specialization argument excludes all relevant rank-one Lang twists and yields square-root cancellation in its mixed moment. We also estimate the prescribed-order population at every nonzero relative trace, uniformly over the intermediate field, and classify all empty slices: the subgroup of order 9 at $s = 3$ and the subgroup of order 11 at $s = 5$. The counts determine scalar realizations of an affine complete mapping and its prescribed companion. General index bounds and a norm obstruction locate the optimal uniform index exponent at $1/4$, with a logarithmic gap in the sufficient range.

1 Introduction

Multiplicative order conditions become linked when two field elements differ by one. Requiring a and $1+a$ to generate a prescribed subgroup is weaker than requiring each to generate it, and the two conditions lead to different densities. For the subgroup C_d of order $d > 1$ in a field $\mathbb{F}_q$ of characteristic two, put

$$\mathcal{R}_d = \{a \in C_d : 1 + a \in C_d,\ \langle a, 1+a\rangle = C_d\}, \qquad D_d = \#\{a : \mathrm{ord}(a) = \mathrm{ord}(1+a) = d\}.$$

Our central case is the norm-one group in a quartic tower:

$$t = 2^s\ (s \geq 2), \qquad K = \mathbb{F}_t \subset E = \mathbb{F}_{t^2} \subset L = \mathbb{F}_{t^4}, \qquad d = (t+1)(t^2+1).$$

Here $C_d = \ker N_{L/K}$ has growing index $t-1$ in L^*. The full intersection has order of magnitude t^2, so an ambient-field character error of square-root size t^2 does not resolve its density. We use norm fibers over the smaller field K to recover the cancellation needed for effective counts. The same geometry controls individual relative traces, where the populations have size $O(t)$ and empty slices require a separate analysis.

Main results

Write $A = t + 1$, $B = t^2 + 1$, and

$$\theta(d) = \frac{\varphi(d)}{d}, \qquad \theta_2(d) = \prod_{p|d}(1 - p^{-2}), \qquad W(d) = 2^{\omega(d)}.$$

The main results form five groups.

Norm-fiber densities. Theorems 4.3 and 4.4 give explicit fixed-fiber and full-family estimates. In particular,

$$|\mathcal{R}_d| = \theta_2(d)(t+1)^2 + O(t^{3/2}W(d)), \qquad D_d \sim \theta(d)^2(t+1)^2.$$

The effective double full-order error is stated separately in (4.5). The local factors reflect the distinction between the two populations: joint generation excludes simultaneous membership in a maximal subgroup, giving $1 - p^{-2}$, whereas full order for each element gives $(1 - p^{-1})^2$.

Complete primitive inner-norm support. For $b \in U = \ker N_{E/K}$, let $D_{d,b}$ count the double full-order pairs with $N_{L/E}(a) = b$. Theorem 4.5 proves, for every $s \geq 2$,

$$D_{d,b} > 0 \quad \Longleftrightarrow \quad \operatorname{ord}(b) = t + 1.$$

Every nonempty such fiber contains at least two elements. The necessity is elementary; sufficiency requires a prime sieve on the three varying order coordinates, an infinite tail bound and complete finite norm coverage. It includes composite $B = t^2 + 1$ and removes all small-parameter restrictions from primitive inner-norm positivity. It prescribes one inner norm, without fixing the second inner norm of $1 + a$.

Distribution across primitive inner norms. Theorem 4.7 expresses the unsieved population N_b as $t + 1 + \mathrm{Kl}_t((1 + b + b^{-1})^{-1})$ and proves $\mathrm{Var}_{\mathcal{P}}(N_b) \sim t$ on the primitive inner norms. For odd s, Theorem 4.8 identifies the cubic coordinate $R_{3,b}$ of the second-norm sieve with a genus-two trace and gives

$$|\mathbb{E}_{\mathcal{P}}[(N_b - (t+1))R_{3,b}]| \leq C_0 W(t+1)\sqrt{t}.$$

A projective Frobenius-spectrum criterion and an exact $\mathbb{F}_8$ certificate exclude every relevant rank-one Lang twist. The fixed geometric constant C_0 is not evaluated numerically. The resulting full-order variance bounds improve constants, while retaining their $W(A)^2$ factor; the cubic coordinate is a single part of the complete order sieve.

All nonzero relative traces and their exceptional groups. For $c \in E$ define

$$M_c = \#\{a \in C_d : \operatorname{ord}(a) = d,\ a + a^{t^2} = c\}.$$

Theorem 5.1 proves

$$|M_c - \theta(d)(t+1)| \leq \theta(d)(4W(d) - 2)\sqrt{t} \qquad (c \in E^*).$$

Theorem 5.4 completes this quantitative result by classifying every empty nonzero slice:

$$M_c = 0 \quad \Longleftrightarrow \quad (s = 3 \text{ and } c^9 = 1) \quad \text{or} \quad (s = 5 \text{ and } c^{11} = 1) \qquad (c \in E^*).$$

Thus the exceptional traces form C_9 at $t = 8$ and C_{11} at $t = 32$. Their different support obstructions are explained by a common inverse-support reduction. Within each fixed-norm class of traces, a cyclic convolution gives the exact mean and a variance bound without the $W(A)$ factor of the pointwise sieve. Scaling transfers the classification to arbitrary nonzero K-norms, including ambient primitive elements when the prescribed norm is primitive (Corollary 5.7).

For $c = 1$, the relation $1 + a = a^{t^2}$ couples the two orders. The exceptional groups each meet K^* only at 1, so this trace-one subfamily is positive exactly when $s \notin \{3, 5\}$. Two exact certificates outside that subfamily cover $s = 3, 5$, giving a retained alternative proof that $D_d > 0$ for all $s \geq 2$. The complete inner-norm support theorem proves this existence independently and with the first inner norm prescribed.

Affine realizations and the uniform index exponent. The arithmetic counts have a direct group-theoretic interpretation. For a complete mapping f, write $F(f) = (\mathrm{id} + f)^{-1}$. Theorem 2.3 proves, for every affine complete mapping $f(x) = Mx + v$ on $\mathbb{F}_2^n$,

$$\langle f, F(f) \rangle = \mathbb{F}_2[M]v \rtimes \langle M, (I + M)^{-1} \rangle.$$

This includes reducible and nonsemisimple M and the vector $v = 0$. When $q = 2^n$ and $\mathrm{ord}_d(2) = n$, precisely $(q-1)|\mathcal{R}_d|$ scalar affine maps realize the prescribed group $\mathbb{F}_q^+ \rtimes C_d$ in fixed field coordinates. For admissible indices $k = (q-1)/d > 1$, the general sufficient range is

$$k \leq \frac{q^{1/4}}{\sqrt{\log_2 q}} \qquad (n \geq 16).$$

The quartic family is realizable at $k = q^{1/4} - 1$, while a norm obstruction gives nonrealizable primitive groups at $k = q^{1/4} + 1$. These results determine the optimal uniform exponent $1/4$. A factor $\sqrt{\log_2 q}$ remains between the sufficient range and the obstructed family.

Why the fibers give stronger estimates

Fixing $b = N_{L/E}(a)$ in $U = \ker N_{E/K}$ isolates a curve over K. For $b^3 \neq 1$ it has genus one. Characters induced by the fixed norm can be constant even when nontrivial; separating these exceptions leaves an $8\sqrt{t}$ mixed-character estimate. The local joint-generation coefficient varies with b. Averaging it exactly, rather than replacing it by a fixed-fiber value, produces $\theta_2(d)$ in the full-family main term. The bad fibers $b^3 = 1$ are treated separately. Every primitive b satisfies $b^3 \neq 1$. For those fibers, the fixed U-component already has full order; only $w \in U$ and the two V-components remain to be sieved. The complete support theorem closes this sieve for all parameters, using exact integer bounds and witnesses for every remaining Frobenius orbit of primitive norms.

For the trace problem, write $c = hu$ with $h \in K^*$ and $u \in U$. The reciprocal-trace substitution $a = c/(1 + \gamma)$ leads, up to a character factor of absolute value one, to a genus-one curve with four geometric boundary points. Its $4\sqrt{t}$ character estimate is uniform on E^*. A prime sieve gives positivity in the large range; exact support certificates settle the remaining fields. In that finite range A has at most two distinct prime divisors, and the inverse-support lemma requires both this restriction and the stated separation condition. The independent integer tail bound closes the infinite range. These parts together prove the complete classification.

Relation to earlier work

The affine companion construction originated in Li's earlier single-author note on the six-dimensional Singer example [31]. The present jointly authored article develops the norm-fiber, relative-trace and

distribution theory beyond that focused note. The earlier note retains its own title, DOI and sole authorship; its record is distinct from the version chain of this article.

The counting objects have direct predecessors. The statistic D_d is the $(t-1)$-primitive-pair count of [13, Proposition 3.1] and a specialization of the norm-constrained count in [14, Definition 3.1]. The populations M_c occur in prescribed-order/trace and line-count frameworks [22, 23, 10, 6]; M_1 is $N_{t^4}(B, A)$ of [17, Theorem 1.2]. The leading double full-order density is already represented by the earlier ambient-field estimate. The quantitative advance here is the effective fiber error in this growing-index family, its norm-dependent coefficients and their exact average, together with the uniform relative-trace estimate and complete exceptional groups. The complete inner-norm support theorem adds positivity at every specified primitive norm and every parameter; this is stronger than full-family existence and is a separate closure of the existing geometric estimate and classical prime sieve.

We use classical freeness indicators and prescribed-order sieves [7, 8, 9, 13], the curve-character theorem and Lang construction [5, 4], and the reciprocal-trace change of variables [17, §4]. The family-specific work verifies the descended curves, constant-character exceptions and ramification data that make these tools effective on the individual fibers. The added moment calculation uses Katz's classical Kloosterman monodromy and cohomological weight bounds [29]. The cubic reduction applies the classical Dickson substitution and Artin–Schreier genus calculation to this specific sieve coordinate. Its mixed-moment proof supplies the required twisted-component exclusion, including a finite algebraic spectrum certificate; the proof of the general representation criterion is included.

The complete-mapping framework comes from Latin squares [1]; triality is the cyclic coordinate-permutation symmetry of a transversal [2]. The prescribed companion pair studied here differs from larger collections of complete mappings [18, 19].

Appendix E and Supplementary Appendix S.C give exact parameter substitutions, including the stronger prime-sieve and rational-image criteria [25, 26, 27]. Their comparisons concern the stated bounds and quantifiers. They also distinguish a specified trace from primitive-normal aggregates [11] and projective norm-one geometry [12], credit the known $s = 2$ positivity case, and discuss related cyclotomic and norm/trace results [15, 20, 21]. Detailed comparisons remain available there without interrupting the proof sequence.

Organization

We begin with affine companion groups because their scalar realization problem identifies $\mathcal{R}_d$. Section 2 gives this reduction and the exact sieve; Sections 3–5 develop the norm-fiber estimates, densities, complete inner-norm support, inner-norm distributions and relative-trace classification. Section 6 returns to the realization range and its obstruction. The appendices contain the necessary certificates, verification scope and relative-trace comparisons. The supplement gives coordinate Singer examples, oriented spreads, nonlinear companions and further scalar statistics, together with the alternative trace-one proof, extended certificates and detailed verification records. Its numbers carry an S prefix.

2 Affine companions and the scalar realization problem

2.1 Complete mappings and their companions

Let $V = \mathbb{F}_2^n$, with addition denoted by $+$ or $\oplus$. A permutation f is a *complete mapping* if $(\mathrm{id} \oplus f)(x) = x \oplus f(x)$ is also a permutation. Set $F(f) = (\mathrm{id} \oplus f)^{-1}$. Function powers denote composition; orders of field elements are multiplicative orders. Composition is read from right to

left.

Lemma 2.1. *Every complete mapping f of an additive group of exponent two has exactly one fixed point, namely $F(f)(0)$.*

Proof. The equation $f(x) = x$ is equivalent to $x \oplus f(x) = 0$. Since $\mathrm{id} \oplus f$ is bijective, its unique solution is $F(f)(0)$. □

Lemma 2.2. *On complete mappings of any additive group of exponent two, F is defined at every iterate and satisfies*

$$F^2(f) = \mathrm{id} \oplus f^{-1}, \qquad F^3(f) = f.$$

Proof. Put $g = (\mathrm{id} \oplus f)^{-1}$. The identity $x = g(x) \oplus f(g(x))$ gives

$$\mathrm{id} \oplus g = f \circ g.$$

Thus g is complete. Taking inverses yields

$$F^2(f) = (f \circ g)^{-1} = g^{-1} \circ f^{-1} = (\mathrm{id} \oplus f) \circ f^{-1} = \mathrm{id} \oplus f^{-1}.$$

Consequently $\mathrm{id} \oplus F^2(f) = f^{-1}$ and $F^3(f) = f$. Only addition of exponent two and bijectivity were used. □

The transversal $\{(x, f(x), x + f(x)) : x \in V\}$ is unchanged as a family when its coordinates are permuted. Its cyclic rotation induces F; Supplementary Proposition S3.1 gives this classical interpretation explicitly. For a scalar affine map $f(x) = \alpha x \oplus c$, put $\beta = (1 \oplus \alpha)^{-1}$ and $\kappa = 1 \oplus \alpha^{-1}$. Then $F(f)(x) = \beta x \oplus \beta c$ and $F^2(f)(x) = \kappa x \oplus \alpha^{-1} c$, so the slopes follow $\alpha \mapsto \beta \mapsto \kappa \mapsto \alpha$, with $\alpha\beta\kappa = 1$. The exponent-two hypothesis matters: over $\mathbb{F}_5$, replacing $\oplus$ by ordinary field addition $+$, all three iterates of $f(x) = 2x + 1$ are defined, but $F^3(f)(x) = 2x + 2 \neq f(x)$.

2.2 The generated affine group

For a vector u write $\tau_u(x) = x \oplus u$. If $U \leq V$ is invariant under $M \leq \mathrm{GL}(V)$, we realize $U \rtimes M$ as $\{x \mapsto Lx \oplus u : L \in M,\ u \in U\}$ in the given coordinates. Composition has the usual rule $(u, L)(v, N) = (u \oplus Lv, LN)$.

Theorem 2.3 (Affine companion group)**.** *Let $V = \mathbb{F}_2^n$, $n \geq 1$, and let $f(x) = Ax \oplus c$ be complete, so $A, I + A \in \mathrm{GL}(n, 2)$. Put*

$$B = (I + A)^{-1}, \qquad R = \mathbb{F}_2[A], \qquad M_A = \langle A, B \rangle,$$
$$U = Rc = \mathrm{span}_{\mathbb{F}_2}\{c, Ac, \ldots, A^{n-1}c\}.$$

For every $c \in V$, including $c = 0$,

$$\langle f, F(f) \rangle = U \rtimes M_A, \qquad |\langle f, F(f) \rangle| = 2^{\dim U} |M_A|.$$

In particular its pure translations are exactly $\{\tau_u : u \in U\}$.

Proof. The minimal polynomial of A is coprime to t and $t + 1$. Bezout identities therefore show that A^{-1} and B belong to R; their inverses do as well. Thus $M_A \leq R^\times$ is abelian and preserves U. Cayley–Hamilton gives the displayed finite spanning set for U. Since $g = F(f)$ has the explicit form $g(x) = Bx \oplus Bc$, both generators lie in $U \rtimes M_A$, proving one inclusion.

Using $AB = I + B$ and $[f, g] = f \circ g \circ f^{-1} \circ g^{-1}$, direct composition gives

$$g \circ f(x) = ABx, \qquad f \circ g(x) = ABx \oplus Bc, \qquad [f, g] = \tau_{Bc}.$$

Conjugation by f sends τ_u to τ_{Au}. Hence the generated group contains all translations whose vectors lie in $\operatorname{span}\{A^j Bc : j \geq 0\} = RBc = Rc = U$, where B is a unit of R. It contains $\tau_c \circ f = A$ and $\tau_{Bc} \circ g = B$ as pure linear maps, so it contains M_A as well. This proves the reverse inclusion. Distinct pairs (u, L) give distinct affine maps, yielding the order and translation statements. □

Corollary 2.4 (Action criteria)**.** *In Theorem 2.3, write $H = \langle f, F(f) \rangle$. Then H is transitive on V if and only if $U = V$, equivalently c is a cyclic vector for A. It is primitive if and only if $c \neq 0$ and the characteristic polynomial of A is irreducible. If A is identified with multiplication by $\alpha \in \mathbb{F}_{2^n}$ in this irreducible case, then for $c \neq 0$*

$$|H| = 2^n \operatorname{lcm}\big(\operatorname{ord}(\alpha), \operatorname{ord}(1 \oplus \alpha)\big).$$

It is doubly transitive if and only if $c \neq 0$, A is irreducible, and this least common multiple is $2^n - 1$. In that case $H = \mathrm{AGL}(1, 2^n)$ in these field coordinates.

Proof. The orbit of 0 is exactly U, giving transitivity. When $U = V$, the group contains the full regular translation group and its stabilizer of 0 is M_A. A block containing 0 is an additive subspace: if u belongs to the block, its translate by u meets the block at u, so the block property makes that translate equal to the block. The block is also M_A-invariant. Conversely the cosets of any M_A-invariant subspace form a block system (cf. [16]). An A-invariant subspace is invariant under the polynomial maps B, A^{-1}, B^{-1}, so the M_A-invariant subspaces are exactly the A-invariant ones. The action is therefore primitive precisely when A acts irreducibly; this is equivalent to an irreducible characteristic polynomial. Such an A makes every nonzero vector cyclic, which explains the condition $c \neq 0$.

An irreducible A identifies V with $\mathbb{F}_{2^n}$ and A with multiplication by α; the two multipliers generate $\langle \alpha, (1 \oplus \alpha)^{-1} \rangle$ in the cyclic multiplicative group. Its order is the least common multiple of the two element orders. Double transitivity means that M_A is transitive on $V \setminus \{0\}$, which holds exactly when this subgroup is the entire multiplicative group. Double transitivity implies primitivity, so the conditions are also necessary outside the irreducible case. □

2.3 Scalar realizations and an exact arithmetic sieve

Corollary 2.5 (Realizing prescribed scalar affine subgroups)**.** *Fix $n \geq 2$, put $q = 2^n$, and regard $\mathbb{F}_q$ as an n-dimensional vector space over $\mathbb{F}_2$. For a divisor $d > 1$ of $q - 1$, let C_d be the unique subgroup of order d of $\mathbb{F}_q^\times$, and define*

$$\mathcal{R}_d = \{a \in C_d : \ 1 + a \in C_d, \ \langle a, 1 + a \rangle = C_d\}.$$

The particular affine subgroup $H_d = \mathbb{F}_q^+ \rtimes C_d \leq \mathrm{AGL}(1, q)$ is generated by f and $F(f)$ for a scalar affine complete mapping $f(x) = ax + c$ if and only if

$$\operatorname{ord}_d(2) = n \quad \textit{and} \quad \mathcal{R}_d \neq \varnothing.$$

Here $\operatorname{ord}_d(2)$ denotes the multiplicative order of 2 modulo the odd integer d. When the conditions hold, the exact number of scalar affine maps f realizing this specified subgroup H_d (in these fixed field coordinates) is

$$(q - 1)|\mathcal{R}_d|.$$

Proof. If $\langle f, F(f)\rangle = H_d$, Theorem 2.3 shows that its linear part is $\langle a, (1+a)^{-1}\rangle = C_d$, so $a \in \mathcal{R}_d$. Its pure translations fill the field, which forces $c \neq 0$ and $\mathbb{F}_2[a] = \mathbb{F}_q$. Since $C_d \leq \mathbb{F}_q^\times$ and $\langle a, 1+a\rangle = C_d$, the least subfield containing C_d is $\mathbb{F}_2[a]$. The subfield generated by a multiplicative subgroup of order d has degree $\mathrm{ord}_d(2)$ over $\mathbb{F}_2$; thus this order is n.

Conversely, suppose $\mathrm{ord}_d(2) = n$ and $a \in \mathcal{R}_d$. Because $1 + a \in \mathbb{F}_2[a]$ and $C_d = \langle a, 1 + a\rangle$, the field $\mathbb{F}_2[a]$ contains C_d, hence equals $\mathbb{F}_q$. In particular $a \neq 0, 1$, multiplication by a is irreducible over $\mathbb{F}_2$, and for any $c \neq 0$ one has $\mathbb{F}_2[a]c = \mathbb{F}_q$. Theorem 2.3 now gives $\langle f, F(f)\rangle = H_d$. There are exactly $|\mathcal{R}_d|$ admissible slopes and $q - 1$ admissible translations; different pairs (a, c) define different maps. □

This criterion concerns prescribed scalar affine subgroups in fixed field coordinates. It reduces their realizability to the arithmetic question of determining the nonemptiness and size of $\mathcal{R}_d$ uniformly in n, d.

Corollary 2.6 (An arithmetic sieve for realizable slopes)**.** *Retain the notation of Corollary 2.5, and for every divisor e of $q - 1$ set*

$$T_e = |\{a \in C_e : 1 + a \in C_e\}|.$$

Then the number of slopes realizing C_d satisfies

$$|\mathcal{R}_d| = \sum_{e|d} \mu(d/e) T_e, \tag{2.1}$$

where μ is the number-theoretic Moebius function. In fact,

$$T_e = \deg \gcd_{\mathbb{F}_2[t]}\big(t^e + 1, (t + 1)^e + 1\big), \tag{2.2}$$

so these numbers depend on e, not on the ambient field containing C_e. Moreover, for $d > 3$ one has

$$|\mathcal{R}_d| \equiv 0 \pmod 3;$$

for $d = 3$ (necessarily n even), $|\mathcal{R}_3| = 2$. In particular, the full-translation subgroup $\mathbb{F}_{2^n}^+ \rtimes C_3$ is realizable by a scalar affine companion pair in fixed field coordinates exactly when $n = 2$.

Proof. Every pair $(a, 1 + a)$ counted by T_e generates a unique subgroup $C_s \leq C_e$. Therefore $T_e = \sum_{s|e} |\mathcal{R}_s|$, extending the definition by $\mathcal{R}_1 = \varnothing$. Moebius inversion yields (2.1). Since $e \mid q - 1$, both polynomials in (2.2) split into simple linear factors over $\mathbb{F}_q$ (their derivatives are nonzero, as e is odd). Their common roots are precisely the elements a counted by T_e, proving the polynomial-gcd formula.

The fractional-linear transformation $\varphi(a) = (1 + a)^{-1}$ preserves $\mathcal{R}_d$: indeed $\varphi(a)$ and $1 + \varphi(a) = a/(1 + a)$ belong to C_d and generate exactly the subgroup generated by a and $1 + a$. Direct simplification gives $\varphi^3(a) = a$. A fixed point obeys $a^2 + a + 1 = 0$, hence has multiplicative order three and can lie in $\mathcal{R}_d$ only when $d = 3$. Thus for $d > 3$ all φ-orbits in $\mathcal{R}_d$ have length three. For $d = 3$, the two nonidentity elements of $\mathbb{F}_4^\times$ comprise $\mathcal{R}_3$. Finally $\mathrm{ord}_3(2) = 2$, so Corollary 2.5 gives the last claim. □

As a symbolic nonrealizability example in dimension six,

$$\gcd(t^9 + 1, (t + 1)^9 + 1) = t^2 + t + 1,$$

while the analogous gcd for exponent 21 is

$$(t^2+t+1)(t^3+t+1)(t^3+t^2+1).$$

Together with the divisor-sum formula, these identities give $|\mathcal{R}_9| = 2-2 = 0$ and $|\mathcal{R}_{21}| = 8-6-2 = 0$. Although $\mathrm{ord}_9(2) = \mathrm{ord}_{21}(2) = 6$, neither $\mathbb{F}_{64}^+ \rtimes C_9$ nor $\mathbb{F}_{64}^+ \rtimes C_{21}$ can be realized by a scalar affine companion pair, as the displayed polynomial identities show.

The quantities T_e are standard cyclotomic intersection counts. For example, if $k = (q-1)/e$ and χ is a multiplicative character of order k on $\mathbb{F}_q$, extending all powers (including the trivial character) by zero at zero, then

$$T_e = \frac{1}{k^2} \sum_{i,j=0}^{k-1} \sum_{x \in \mathbb{F}_q} \chi^i(x)\chi^j(1+x).$$

This identity is simply the multiplicative-character indicator of C_e. Accounting separately for the trivial-character terms gives the effective classical estimate in Proposition 6.1; its combination with a maximal-subgroup sieve will prove that $\mathcal{R}_d$ is nonempty in the families below.

3 Genus-one norm fibers and mixed characters

3.1 The quartic tower and its fibers

Throughout Sections 3–5, put

$$t = 2^s \ (s \geq 2), \quad K = \mathbb{F}_t, \quad E = \mathbb{F}_{t^2}, \quad L = \mathbb{F}_{t^4}, \quad A = t+1, \quad B = t^2+1, \quad d = AB.$$

Here $\gcd(A,B) = 1$ and $C_d = \ker N_{L/K}$. Write $U = \ker N_{E/K} = C_A$ and $V = \ker N_{L/E} = C_B$; in these sections V denotes this cyclic norm-one group. The ambient multiplicative index is $(t^4-1)/d = t-1$. The aim is to count joint generation and, separately, simultaneous full order for a and $1+a$. We first expose the norm fibers, separate the constant-character exceptions, and estimate the remaining characters on the good fibers $b^3 \neq 1$. Section 4 then combines this geometric input with the two order indicators and averages over the norm. The following elementary quadratic reduction is the characteristic-two norm root-count calculation; compare [15, Proposition 3.6] with base field E. Its role is to expose the fibers to be estimated.

Proposition 3.1 (Quadratic parametrization)**.** *The elements $a \in L \setminus E$ with $a, 1+a \in C_d$ correspond, two at a time, to the ordered pairs $(b,w) \in U^2$ satisfying*

$$c = 1+b+w \neq 0, \qquad \mathrm{Tr}_{E/\mathbb{F}_2}(b/c^2) = 1.$$

The two elements are the roots of X^2+cX+b, and $N_{L/E}(a) = b$, $N_{L/E}(1+a) = w$. The remaining elements, in E, are precisely the two roots of X^2+X+1 when s is odd, and there are none when s is even. These remaining elements generate only C_3.

Proof. If $a \notin E$, write $b = a^{t^2+1}$ and $c = a+a^{t^2}$. Then $c \neq 0$, the minimal polynomial is X^2+cX+b, and $(1+a)^{t^2+1} = 1+b+c = w$. Norm transitivity puts b, w in U. After division by c^2, the quadratic is an Artin–Schreier equation; it is irreducible exactly when the displayed absolute trace is one. Conversely, that trace condition makes the quadratic irreducible over E, with both roots in $L \setminus E$. Their two relative norms are b, w, so norm transitivity gives $a, 1+a \in C_d$. The relative norms determine the ordered pair uniquely.

For $a \in E$, the same norm conditions reduce to $a, 1+a \in U$, since $N_{L/K}(a) = N_{E/K}(a)^2$. Thus $a^t = a^{-1}$ and $(1+a)^t = (1+a)^{-1}$. Substitution gives $(1+a)(1+a^{-1}) = 1$, or $a^2 + a + 1 = 0$. Such roots lie in U exactly when $3 \mid t+1$, equivalently when s is odd. Their two norms satisfy $1 + b + w = 0$. In particular, the omitted $c = 0$ branch contributes no realizing slope. □

Lemma 3.2 (A genus-one norm fiber)**.** *Fix $b \in U$ with $b^3 \neq 1$ and put*

$$X_b(K) = \{a \in L : N_{L/E}(a) = b,\ N_{L/K}(1+a) = 1\}.$$

This set is the full set of K-points of a smooth projective geometrically connected curve C_b/K of genus one. Over $\overline{K}$ the affine part has equation

$$(x+1)(x+b)(y+1)(y+b^{-1}) = xy \qquad (x, y \neq 0). \tag{3.1}$$

Its completion in $\mathbb{P}^1 \times \mathbb{P}^1$ has eight boundary points, none rational over K. Consequently

$$\left|\#X_b(K) - (t+1)\right| \leq 2\sqrt{t}.$$

Proof. The relative-norm equation defines a variety over K by restriction of scalars. In the four split coordinates of $\mathrm{Res}_{L/K}\,\mathbb{G}_m$, write $(a_0, a_1, a_2, a_3) = (x, y, b/x, b^{-1}/y)$. The remaining norm equation is exactly (3.1). Let $P = (x+1)(x+b)$ and $Q = (y+1)(y+b^{-1})$. At a singular point in the torus, $PQ = xy \neq 0$ and

$$(b+1)Q + y = 0, \qquad (b^{-1}+1)P + x = 0.$$

Multiplication would force $b + b^{-1} = 1$, hence $b^3 = 1$. This contradicts the hypothesis $b^3 \neq 1$. The boundary points are

$$D_x = \{(0,1), (0,b^{-1}), (\infty,1), (\infty,b^{-1})\},$$
$$D_y = \{(1,0), (b,0), (1,\infty), (b,\infty)\}.$$

They are distinct and smooth: in the appropriate affine charts the derivative along the finite coordinate is a nonzero multiple of $1+b$ or $1+b^{-1}$. No corner of the square is on the curve. The smooth bidegree-$(2,2)$ curve is geometrically connected: distinct components with zero intersection would all have to be fibers of the same projection, contrary to the total bidegree. It is therefore geometrically irreducible and has genus $(2-1)(2-1) = 1$. Its smooth completion descends to K.

The t-power Frobenius on these coordinates is $(x, y) \mapsto (b/y^t, x^t)$. A rational point satisfies $y = x^t$, which excludes all eight boundary points. Hence $C_b(K) = X_b(K)$, with the stated interpretation in L. The final bound is the Hasse–Weil bound. □

3.2 Character components and the constant exception

We now retain all three varying character components on a norm fiber. For $a \in X_b(K)$ put

$$w = (1+a)^B \in U, \qquad v = a^A \in V, \qquad v_+ = (1+a)^A \in V. \tag{3.2}$$

Lemma 3.3 (Character decomposition)**.** *The map $C_d \longrightarrow U \times V$, $z \mapsto (z^B, z^A)$, is an isomorphism. Every pair $\xi, \zeta \in \widehat{C_d}$ has a unique decomposition*

$$\xi(z) = \alpha(z^B)\chi(z^A), \qquad \zeta(z) = \eta(z^B)\psi(z^A), \quad \alpha, \eta \in \widehat{U}, \quad \chi, \psi \in \widehat{V}.$$

Define the two-character sum and the norm-character exception set by

$$S_b(\xi, \zeta) = \sum_{a \in X_b(K)} \xi(a)\zeta(1+a),$$
$$\mathcal{E}_{\mathrm{norm}} = \{(\alpha \circ (z \mapsto z^B), 1) : \alpha \in \widehat{U}\}. \tag{3.3}$$

For every $b \in U$ the sum is

$$S_b(\xi,\zeta) = \alpha(b) \sum_{a \in X_b(K)} \eta(w)\chi(v)\psi(v_+).$$

Moreover $(\xi,\zeta) \in \mathcal{E}_{\mathrm{norm}}$ if and only if $\eta = \chi = \psi = 1$.

Proof. The two powers have orders dividing A and B, respectively. An element in the kernel has both its A-th and B-th powers equal to one, hence is one because $\gcd(A,B) = 1$. The source and target have the same order AB, so the map is an isomorphism. Taking characters gives the unique decomposition. On the fiber $a^B = b$; substitution gives the identity for the sum. Uniqueness also proves the stated equivalence for the exception set. □

3.3 The mixed-character estimate

The required cancellation is supplied by the following proposition. Its proof keeps the descended Lang equations and their Frobenius action explicit: a geometric Kummer equation alone would not identify the finite-field character trace. There are three steps: identify the descended character traces, use boundary inertia to rule out geometric constant components, and apply the tame conductor bound. This keeps the hypotheses of the classical estimate separate from the cyclic-group sieve.

Proposition 3.4 (Mixed characters on a norm fiber)**.** *Let $b \in U$ satisfy $b^3 \neq 1$. For characters $\eta \in \widehat{U}$ and $\chi, \psi \in \widehat{V}$, not all trivial,*

$$\left| \sum_{a \in X_b(K)} \eta(w)\chi(v)\psi(v_+) \right| \leq 8\sqrt{t}. \tag{3.4}$$

The product Lang cover used for this sum is geometrically connected, and hence its smooth projective normalization is geometrically irreducible. Every character quotient is tame, with conductor degree at most eight.

Proof. Let $C_b^\circ = C_b \setminus (D_x \cup D_y)$, with the eight points of Lemma 3.2. This open curve has $C_b^\circ(K) = X_b(K)$. Write $\sigma \in \mathrm{Gal}(L/K)$ for $z \mapsto z^t$, extended as $\sigma \otimes 1$ on every $L \otimes_K R$. The algebraic map

$$\nu(z) = z\,\sigma(z)$$

sends the norm-one torus $\ker N_{L/K}$ to $\mathcal{T} = \mathrm{Res}_{E/K}(\ker N_{L/E})$:

$$N_{L/E}(\nu(z)) = N_{L/E}(z)\,\sigma(N_{L/E}(z)) = N_{L/K}(z) = 1.$$

Its value on K-points is z^A. This construction is needed for the varying norm of $1+a$: exponentiation by A on geometric points would give relative norm w^A, which is not the constant function one.

Set $w_0 = (x+1)(x+b)/x$. The split coordinates of the three K-defined maps to the norm tori are

	first coordinate	second coordinate
$N_{L/E}(1+a)$	w_0	w_0^{-1}
$\nu(a)$	xy	by/x
$\nu(1+a)$	$(x+1)(y+1)$	$(y+1)(x+b)/x$

The first row is a map to $\mathcal{U} = \ker N_{E/K}$, and $\mathcal{U}(K) = U$, $\mathcal{T}(K) = V$. On $\mathcal{U}$ the Frobenius is $z \mapsto z^{-t}$; its Lang equation is $z^{-A} = w_0$. On $\mathcal{T}$ the Frobenius is $(z_0, z_1) \mapsto (z_1^{-t}, z_0^t)$; the Lang equations $z_1^{-t}/z_0 = r_0$, $z_0^t/z_1 = r_1$ give $z_0^B = r_1^t/r_0$. Consequently the product pullback is, over $\overline{K}$,

$$z_U^A = w_0^{-1}, \qquad z_1^B = g_1, \qquad z_2^B = g_2, \tag{3.5}$$

where

$$g_1 = b^{-1}\frac{y^{t-1}}{x^A}, \qquad g_2 = w_0^t\frac{(y+1)^{t-1}}{(x+1)^A}. \tag{3.6}$$

For the second formula, substitute $x + b = xw_0/(x+1)$ in r_1^t/r_0.

These geometric equations inherit their K-structure from the Lang pullbacks. More explicitly, above $F_{C_b}(x,y) = (b/y^t, x^t)$ their Frobenius sends the three displayed covering coordinates to

$$(w_0 z_U,\ xyz_1,\ (x+1)(y+1)z_2). \tag{3.7}$$

This follows from $F(z) = rz$ in the Lang equations. It is compatible with (3.5), since

$$w_0(F_{C_b}) = w_0^{-t}, \quad g_1(F_{C_b}) = (xy)^B g_1, \quad g_2(F_{C_b}) = ((x+1)(y+1))^B g_2.$$

At a rational point $(x,y) = (a,a^t)$, the three deck multipliers in (3.7) are precisely w, v, v_+. Thus the character component has trace $\eta(w)\chi(v)\psi(v_+)$; the inverse Frobenius convention simultaneously inverts the characters and gives the same bound.

We check geometric connectedness and the conductor directly. The divisors of w_0, g_1, g_2 are supported on the eight boundary points. Their valuations are

P	$\mathrm{ord}_P(w_0)$	$\mathrm{ord}_P(g_1)$	$\mathrm{ord}_P(g_2)$
$(0,1)$	-1	$-A$	-1
$(0,b^{-1})$	-1	$-A$	$-t$
$(\infty,1)$	-1	A	t
(∞,b^{-1})	-1	A	1
$(1,0)$	1	$t-1$	-1
$(b,0)$	1	$t-1$	t
$(1,\infty)$	1	$-(t-1)$	$-t$
(b,∞)	1	$-(t-1)$	1

(3.8)

Indeed x, y have simple zeros and poles on D_x, D_y; $x+1$ has simple zeros at $(1,0),(1,\infty)$ and simple poles at the two points with $x = \infty$, and the corresponding assertion holds for $y+1$. Substitution in (3.6) gives each entry of the table.

For every $m \mid B$, a relation $g_1^r g_2^s = f^m$ in $\overline{K}(C_b)^\times$ would imply, at $(0,1)$ and $(0,b^{-1})$,

$$m \mid Ar + s, \qquad m \mid Ar + ts.$$

Now $\gcd(B, t-1) = \gcd(t^2+1, t-1) = 1$ and $\gcd(A,B) = 1$. Subtracting forces $m \mid s$, and then $m \mid r$. The valuation -1 of w_0 likewise excludes every nontrivial Kummer power relation for an order dividing A. One can verify simultaneous independence in a common exponent:

$$(w_0^{-1})^{Be_0} g_1^{Ae_1} g_2^{Ae_2} \in \overline{K}(C_b)^{\times AB} \quad \Longrightarrow \quad A \mid e_0,\ B \mid e_1,\ B \mid e_2.$$

Reduction of the first two valuations modulo A gives $A \mid e_0$; their difference modulo B gives $B \mid e_2$, and either valuation then gives $B \mid e_1$. This is independence of the relevant functions modulo powers in $\overline{K}(C_b)^\times$, verified by their local valuations.

At $P_1 = (0,1)$ and $P_2 = (0,b^{-1})$, the geometric inertia elements for the Kummer functions (w_0^{-1}, g_1, g_2), written in $G = \mathbb{Z}/A\mathbb{Z} \times \mathbb{Z}/B\mathbb{Z} \times \mathbb{Z}/B\mathbb{Z}$, are

$$v_1 = (1, -A, -1), \qquad v_2 = (1, -A, -t).$$

Their difference is $(0, 0, t-1)$, which generates the third factor because $\gcd(t-1, B) = 1$. Combining a suitable multiple of this element with v_1 gives $(1, -A, 0)$. Its order is AB: the first coordinate has order A, the second has order B since $\gcd(A, B) = 1$. Thus v_1, v_2 generate all of G. Equivalently, no nontrivial character annihilates both inertia elements. The subgroup generated by geometric inertia is therefore the full deck group, proving geometric connectedness of the product cover and of every character quotient.

All covering degrees are odd, so wild inertia is trivial. For a rank-one character its local Artin conductor is one if the inertia character is nontrivial, and zero otherwise. Thus its conductor degree is the number r of active rows of (3.8), and $r \le 8$. Cancellation can occur at individual rows; we do not presume ramification at every boundary point. For a nontrivial character triple the inertia test proves geometric nontriviality and $r > 0$. The genus-one Artin-character Weil bound now gives $(2 \cdot 1 - 2 + r)\sqrt{t} \le 8\sqrt{t}$; see [5, §4, Lemma 1 in the author's version] and the Lang construction in [4, pp. 198–199]. In cohomological terms, extend over the inactive boundary points. On the complement of the r active points, H^0_c, H^2_c vanish and the tame Euler-characteristic formula gives $\dim H^1_c = 2g - 2 + r = r$. The eigenvalues have absolute value at most $\sqrt{t}$, by the Weil theorem on the character component of the finite cover. No active boundary point is K-rational, so this trace sum is exactly the sum over $X_b(K)$, with no rational boundary correction. □

Corollary 3.5 (The precise two-character exception)**.** *Let $b \in U$ satisfy $b^3 \neq 1$ and let $\xi, \zeta \in \widehat{C_d}$. Use the character decomposition lemma and* (3.3)*. If $(\xi, \zeta) \in \mathcal{E}_{\mathrm{norm}}$, then*

$$S_b(\xi, \zeta) = \alpha(b) \# X_b(K).$$

In every other case $|S_b(\xi, \zeta)| \le 8\sqrt{t}$. In particular, the assertion for every nontrivial two-character pair is false: a nontrivial $\alpha \circ N_{L/E}$ with $\zeta = 1$ is constant on the norm fiber.

Proof. The decomposition lemma identifies the exception set with the trivial varying triple. That triple gives the exact constant sum. Every pair outside the set has a nontrivial varying triple, so Proposition 3.4 gives the bound. □

4 Joint generation and double full-order densities

4.1 Two cyclic-group indicators

The mixed-character estimate separates the geometric input from the cyclic-group sieve. Joint generation excludes simultaneous membership in a maximal subgroup; individual full order excludes membership for each element separately. Their local factors and therefore their final densities differ.

Lemma 4.1 (The primitive-element indicator)**.** *For a cyclic group G of order n, put $\theta(n) = \varphi(n)/n$ and $W(n) = 2^{\omega(n)}$. For $z \in G$,*

$$\mathbf{1}_{\mathrm{ord}(z)=n} = \theta(n) \sum_{m \mid n} \frac{\mu(m)}{\varphi(m)} \sum_{\mathrm{ord}(\chi)=m} \chi(z).$$

The constant coefficient is $\theta(n)$, and the total absolute coefficient sum is $\theta(n)W(n)$.

Proof. The character sum factors over the primes $p \mid n$. Its p-factor is $1 - (p-1)^{-1} \sum_{\mathrm{ord}(\chi)=p} \chi(z)$, which is zero on G^p and $p/(p-1)$ off G^p. Multiplication by the local factor $(p-1)/p$ of $\theta(n)$ therefore gives exactly the full-order indicator. Distinct squarefree orders have disjoint character supports; each squarefree divisor contributes absolute coefficient mass one before multiplication by $\theta(n)$. There are $W(n)$ of them. □

Lemma 4.2 (Local averaging in a cyclic group)**.** *Put*

$$\theta_2(n) = \prod_{p|n}(1-p^{-2}), \qquad J_b = \{p \mid A : b \in U^p\},$$
$$\lambda_A(b) = \prod_{p\in J_b}(1-p^{-1}), \quad H(A) = \prod_{p|A}\frac{p+2}{p+1}.$$

The following are exact averages over all A elements of U:

$$\begin{aligned} \frac{1}{A}\sum_{b\in U}\lambda_A(b) &= \theta_2(A), \\ \frac{1}{A}\sum_{b\in U}\lambda_A(b)2^{|J_b|} &= \theta_2(A)H(A). \end{aligned} \tag{4.1}$$

In particular the average of $\lambda_A(b)\theta_2(B)$ is $\theta_2(A)\theta_2(B) = \theta_2(d)$.

Proof. Fix a generator u and write $b = u^j$, $j \in \mathbb{Z}/A\mathbb{Z}$. For $p \mid A$ the condition $b \in U^p$ is equivalent to $p \mid j$. Reduction of uniform j modulo $\mathrm{rad}(A)$ is uniform; the Chinese remainder theorem makes the residues modulo the distinct primes independent. The probability of $p \in J_b$ is therefore $1/p$. The local expectation for λ_A is

$$(1-1/p) + \frac{1}{p}(1-1/p) = 1-p^{-2}.$$

For $\lambda_A(b)2^{|J_b|}$ it is

$$(1-1/p) + \frac{2}{p}(1-1/p) = (1-p^{-2})\frac{p+2}{p+1}.$$

Multiplication of these independent local expectations proves both identities. Since $\gcd(A,B) = 1$, their prime supports are disjoint, which proves the last equality. □

4.2 Joint generation

Theorem 4.3 (Joint generation density at the inner boundary)**.** *Use $\theta_2, \lambda_A, J_b, H(A)$ from the local averaging lemma, and put $W(n) = 2^{\omega(n)}$, $r_0 = \gcd(3,A) \le 3$. Let $\mathcal{R}_{d,b} = \mathcal{R}_d \cap X_b(K)$. For every $b \in U$ with $b^3 \neq 1$,*

$$\big||\mathcal{R}_{d,b}| - \lambda_A(b)\theta_2(B)A\big| \le \lambda_A(b)\theta_2(B)\big(8\,2^{|J_b|}W(B) - 6\big)\sqrt{t}. \tag{4.2}$$

For the full family,

$$\big||\mathcal{R}_d| - \theta_2(d)A^2\big| \le \theta_2(d)A\big(8H(A)W(B) - 6\big)\sqrt{t} + 2r_0A. \tag{4.3}$$

Since $H(A) \le W(A)$, this implies

$$|\mathcal{R}_d| = \theta_2(d)t^2 + O(t^{3/2}W(d)), \qquad |\mathcal{R}_d| \sim \theta_2(d)t^2.$$

The fixed-fiber population is $O(t)$, and its coefficient depends on b. If b is primitive in U, then J_b is empty and the coefficient is $\theta_2(B)$. Moreover $|T_d - A^2| \le 2A\sqrt{t} + r_0A$, so the actual proportion of jointly generating slopes among the ordinary intersection is $|\mathcal{R}_d|/T_d = \theta_2(d) + O(W(d)/\sqrt{t})$.

Proof. The group isomorphism gives

$$\langle a, 1+a\rangle = C_d \quad \Longleftrightarrow \quad \langle b, w\rangle = U \text{ and } \langle v, v_+\rangle = V.$$

For $p \mid A$ with $b \notin U^p$ the first condition is automatic. For $p \in J_b$ it requires $w \notin U^p$. By orthogonality its indicator is the product of the factors

$$1 - \frac{1}{p}\sum_{\eta^p=1} \eta(w), \qquad p \in J_b.$$

Each factor has constant coefficient $1 - 1/p$ and total absolute coefficient sum $2(1 - 1/p)$. The second condition has indicator

$$\prod_{p\mid B}\left(1 - \frac{1}{p^2}\sum_{\chi^p=\psi^p=1} \chi(v)\psi(v_+)\right).$$

Here each factor has constant coefficient $1 - 1/p^2$ and total absolute coefficient sum $2(1 - 1/p^2)$. Characters belonging to different prime-primary components have unique products, so multiplying these expansions gives constant coefficient $c_b = \lambda_A(b)\theta_2(B)$ and total absolute coefficient sum $c_b 2^{|J_b|}W(B)$. The Hasse error for the trivial triple is $2c_b\sqrt{t}$; Proposition 3.4 bounds all other terms by $8\sqrt{t}$. This proves (4.2).

The local averaging lemma supplies (4.1). Multiplication by $\theta_2(B)$ turns $\theta_2(A)$ into $\theta_2(d)$; it also turns the weighted average into $\theta_2(d)H(A)$. These identities average over all $b \in U$, before separating the exceptional fibers.

There are precisely r_0 exceptional b with $b^3 = 1$. For every b, each $w \in U$ allows at most two roots of $X^2 + (1 + b + w)X + b$. Thus $\#X_b(K) \le 2A$ even in an exceptional fiber. Its contribution differs from the formal main term $c_b A \in [0, A]$ by at most $2A$. Sum (4.2) over the good fibers, bound the positive error coefficients by their sum over all b, and use the two exact averages. This gives (4.3). It includes all exceptional and subfield points, rather than applying the smooth-fiber bound to them. Finally $W(d) = d^{o(1)} = t^{o(1)}$, while $\theta_2(d) \ge 1/2$: the latter follows by comparison with the telescoping product $\prod_{n=2}^{\infty}(1 - n^{-2}) = 1/2$. Replacing $A^2 = (t+1)^2$ by t^2 gives the stated asymptotic. For T_d sum the Hasse bound over the good fibers and use $|\#X_b(K) - A| \le A$ on each exceptional fiber. This proves its bound and, by division, the density assertion. □

4.3 Both linked slopes of full order

Theorem 4.4 (Double full-order statistics and existence for every $s \ge 2$)**.** *Define $\theta(n) = \varphi(n)/n$ and*

$$D_{d,b} = \#\{a \in X_b(K) : \operatorname{ord}(a) = \operatorname{ord}(1+a) = d\}, \qquad D_d = \sum_{b\in U} D_{d,b}.$$

If b is not primitive in U, then $D_{d,b} = 0$. For primitive b, put $c = \theta(A)\theta(B)^2$. Then

$$|D_{d,b} - cA| \le c(8W(A)W(B)^2 - 6)\sqrt{t}. \tag{4.4}$$

In particular,

$$|D_d - \theta(d)^2A^2| \le \theta(d)^2A(8W(A)W(B)^2 - 6)\sqrt{t}, \tag{4.5}$$

and $D_d \sim \theta(d)^2t^2$ as $s \to \infty$. Define the explicit analytic range

$$\mathcal{S}_{\text{an}} = \{s \in \mathbb{Z}_{\ge 2} : A > (8W(A)W(B)^2 - 6)\sqrt{t}\}, \tag{4.6}$$

with $t = 2^s$, $A = t + 1$, and $B = t^2 + 1$. For every $s \in \mathcal{S}_{\text{an}}$ and every primitive $b \in U$, $D_{d,b} > 0$. This set contains all sufficiently large integers s. Moreover, $D_d > 0$ for every integer $s \ge 2$.

Proof. Order d forces $a^B = b$ to have order A, which proves the vanishing assertion. For primitive b, the two full-order conditions are exactly that w is primitive in U and v, v_+ are primitive in V. Apply Lemma 4.1 separately to these three variables. For a cyclic group of order n it has constant coefficient $\theta(n)$ and total absolute coefficient sum $\theta(n)W(n)$. The product therefore has constant coefficient c and total absolute coefficient sum $cW(A)W(B)^2$. The trivial triple contributes $c\#X_b(K)$, with Hasse error $2c\sqrt{t}$; every other triple has the mixed bound $8\sqrt{t}$. This proves (4.4). Sum over the $\varphi(A) = A\theta(A)$ primitive norm fibers and use $\theta(d) = \theta(A)\theta(B)$ to obtain (4.5). The relative error tends to zero because $W(A)W(B)^2 = t^{o(1)}$. The stated inequality makes the lower bound positive in every primitive norm fiber. The all-parameter assertion follows from the stronger Theorem 4.5, proved below: every primitive inner norm occurs, so $D_d > 0$ for all $s \geq 2$. That support theorem uses the mixed estimate and a separate prime-sieve closure, independently of the trace classification. □

4.4 Every primitive inner norm occurs

The preceding error estimate gives a sufficient range for each primitive norm. A prime sieve and exact finite closure remove that range restriction. Here and below, primitive in U means order $A = t + 1$, not primitive in the ambient group E^*.

Theorem 4.5 (Complete primitive inner-norm support)**.** *For every integer $s \geq 2$ and every $b \in U$,*

$$D_{d,b} > 0 \quad \Longleftrightarrow \quad \operatorname{ord}(b) = A.$$

In fact $D_{d,b} \geq 2$ for every primitive b. Thus the image of the double full-order population under $N_{L/E}$ is exactly the set of generators of U, including when $B = t^2 + 1$ is composite.

We first give the fixed-fiber sieve used in the proof. An element of a cyclic group is e-free if it belongs to no subgroup of pth powers for any prime $p \mid e$.

Lemma 4.6 (Three-coordinate sieve on a primitive norm fiber)**.** *Fix $s \geq 2$ and* a primitive $b \in U$, *so that* $\operatorname{ord}(b) = A$ *and* $b^3 \neq 1$. *For squarefree $e_A \mid A$ and $e_1, e_2 \mid B$, let $N_b(e_A, e_1, e_2)$ count the points of $X_b(K)$ for which w, v, v_+ in* (3.2) *are respectively e_A, e_1, e_2-free. Put*

$$\Theta = \theta(e_A)\theta(e_1)\theta(e_2), \qquad W_c = 2^{\omega(e_A)+\omega(e_1)+\omega(e_2)}.$$

List the omitted prime positions in the three coordinates, counting a prime twice if omitted in both B coordinates. Let N be their number and set

$$\delta = 1 - \sum_{\text{omitted positions } p} \frac{1}{p}, \qquad \Lambda = 2 + \frac{N-1}{\delta}.$$

If $\delta > 0$, then

$$D_{d,b} \geq \Theta\delta\big[A - (8W_c\Lambda - 6)\sqrt{t}\big]. \tag{4.7}$$

For $N = 0$ the empty sum gives $\delta = 1$ and $\Lambda = 1$.

Proof. Because b is primitive, the fixed U-component of a already has full order. The three remaining conditions are precisely primitivity of w, v, v_+. Lemma 4.1, with its character expansion restricted to divisors of e, and Proposition 3.4 give

$$|N_b(e_A, e_1, e_2) - \Theta A| \leq \Theta(8W_c - 6)\sqrt{t}. \tag{4.8}$$

The trivial character has Hasse error $2\Theta\sqrt{t}$; all other coefficients have total absolute mass $\Theta(W_c - 1)$ and character bound $8\sqrt{t}$.

Write N_c for the core count and N_p for the count with one omitted prime condition added in its specified coordinate. Disjoint prime character supports in that coordinate imply

$$|N_p - \theta(p)N_c| \le 8\Theta\theta(p)W_c\sqrt{t}.$$

Every character in this difference is nontrivial in the augmented coordinate, so the mixed estimate applies. No varying character is absorbed into the fixed norm.

On a core point failing k omitted conditions, the score in $\sum_p N_p - (N-1)N_c$ is $1-k$. It is one if all conditions hold and nonpositive otherwise. This is the classical prime sieve [7], applied to three coordinate positions; it requires no independence of their values. Rewriting its lower bound as

$$\delta N_c + \sum_p (N_p - \theta(p)N_c)$$

and using $\sum_p \theta(p) = N - 1 + \delta$ proves (4.7). If $N = 0$, this is just (4.8). □

Proof of Theorem 4.5. Necessity follows from $\mathrm{ord}(a^B) = A$ when $\mathrm{ord}(a) = AB$. For sufficiency fix a primitive b. Since $A \ge 5$, its fiber satisfies $b^3 \ne 1$ and all the geometric hypotheses of the sieve lemma.

Appendix A proves positivity by the full-core bound for every $s \ge 1924$. For $2 \le s \le 1923$, its exact integer sieve certificates prove positivity for 1909 parameters. The remaining set is

$$\mathcal{E} = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15\}.$$

For these 13 parameters, 1182 exact witnesses cover all 32022 primitive norms by Frobenius orbits. The appendix specifies the complete coverage and the field and order tests. These three ranges exhaust all $s \ge 2$.

Finally $a \mapsto a^{t^2}$ preserves the norm and both orders. It has no fixed point among these witnesses: if $a \in E$ then its order is at most $t^2 - 1 < d$. Thus each nonempty fiber contains at least two such elements. □

The conclusion prescribes the first inner norm b; it does not simultaneously prescribe $w = N_{L/E}(1 + a)$. It strengthens the support statement without changing the errors in (4.4)–(4.5).

4.5 Distribution across primitive inner norms

The support theorem prescribes which inner norms occur. We now study the variation of the populations on those norms. Put

$$\mathcal{P} = \{b \in U : \mathrm{ord}(b) = A\}, \qquad N_b = \#X_b(K), \qquad k_b = N_b - A.$$

All expectations, variances and covariances in this subsection use the uniform probability measure on $\mathcal{P}$. The unsieved population N_b and the double full-order population $D_{d,b}$ are distinct random variables.

For $\lambda \in K^*$ write

$$\mathrm{Kl}_t(\lambda) = \sum_{x \in K^*} (-1)^{\mathrm{Tr}_{K/\mathbb{F}_2}(x + \lambda/x)}.$$

Theorem 4.7 (Unsieved inner-norm moments)**.** *For $b \in U$ with $b^3 \neq 1$, set $\beta_b = (1+b+b^{-1})^{-1} \in K^*$. Then*

$$N_b = A + \mathrm{Kl}_t(\beta_b).$$

For every $s \geq 2$, with $W = W(A)$,

$$\begin{aligned} |\mathbb{E}k_b| &\leq 8Wt/A, \\ \left|\mathbb{E}k_b^2 - t\right| &\leq 15Wt^{3/2}/A, \\ t - 15Wt^{3/2}/A - 64W^2t^2/A^2 \leq \mathrm{Var}(N_b) &\leq t + 15Wt^{3/2}/A. \end{aligned}$$

Consequently $\mathbb{E}N_b = A + O(W(A))$ and

$$\mathrm{Var}(N_b) = t + O\big(W(A)\sqrt{t} + W(A)^2\big) \sim t.$$

The exact Kloosterman parameter and the restriction to primitive b are used together in this result. Its second-moment main term uses the classical SL_2 monodromy theorem for Kl_2, including characteristic two [29, Theorem 11.1]. The proof, with its normalization and five geometric boundary points, is in Supplementary Section S.E.

For odd $s \geq 3$, the factor $3 \mid A$ gives one concrete coordinate of the primitive-w sieve. Let $H(b, w)$ be the root count in Proposition 3.1, and define

$$R_{3,b} = \sum_{w \in U} (31_{w \in U^3} - 1)H(b, w), \qquad P_{3,b} = \sum_{w \notin U^3} H(b, w) = \frac{2N_b - R_{3,b}}{3}.$$

Here $P_{3,b}$ filters only the prime 3 in the second inner norm. The other primes of A and both B-order conditions remain separate.

Theorem 4.8 (The cubic coordinate and its mixed moment)**.** *Suppose $s \geq 3$ is odd and $b \in \mathcal{P}$. Take the unique $u \in U$ with $u^2 = b$, and put $h = u + u^{-1} \in K \setminus \{0, 1\}$. Let C_h be the smooth projective model of*

$$Y^2 + Y = \frac{X^3 + X}{h^2 + 1 + h(X^3 + X)}, \qquad a_h = t + 1 - \#C_h(K). \tag{4.9}$$

Then C_h has genus two and

$$R_{3,b} = -a_h, \qquad |R_{3,b}| \leq 4\sqrt{t}.$$

There is an absolute constant C_0 such that, for all these s,

$$|\mathbb{E}[k_b R_{3,b}]| \leq C_0 W(A) t^{3/2}/A \leq C_0 W(A)\sqrt{t}. \tag{4.10}$$

In particular

$$|\,\mathrm{Cov}(N_b, R_{3,b})| \leq (C_0 + 32)W(A)t^{3/2}/A.$$

The primitive-b assumption is essential to the displayed trace identity: for good $b \in U^3$ the same calculation gives $R_{3,b} = 2a_h$. Odd s is also essential to the cubic coordinate. The proof first applies the classical Dickson substitution $D_3(x + x^{-1}) = x^3 + x^{-3}$ and counts all boundary contributions. For the mixed moment, a projective Frobenius-spectrum criterion excludes every relevant rank-one Lang twist between the rank-two Kloosterman sheaf and the rank-four genus-two sheaf. Supplementary Section S.E.4 gives the general criterion and an exact $\mathbb{F}_8$ certificate. The constant C_0 is defined by the conductor of a fixed family in (S.E.1); no numerical value is asserted.

Corollary 4.9 (Consequences for sieve variances)**.** *For odd $s \geq 3$, put $V_R = \mathrm{Var}(R_{3,b})$. Then $0 \leq V_R \leq 16t$ and*

$$\mathrm{Var}(P_{3,b}) = \frac{4}{9}t + \frac{1}{9}V_R + O(W(A)\sqrt{t} + W(A)^2).$$

For the full-order count, set $c = \theta(A)\theta(B)^2$ and $M = W(A)W(B)^2$. For each primitive b,

$$|D_{d,b} - cA| \leq c(8M - 12)\sqrt{t}. \tag{4.11}$$

Writing $V_N = \mathrm{Var}(N_b)$ also gives

$$\sqrt{\mathrm{Var}(D_{d,b})} \leq c\Big[8(M-2)\sqrt{t} + \sqrt{V_N + 4t + (C_0 + 32)W(A)t^{3/2}/A}\Big].$$

In particular, along odd $s \to \infty$,

$$\mathrm{Var}(D_{d,b}) \leq c^2\big[8(M-2) + \sqrt{5 + o(1)}\big]^2 t.$$

Proof. Expand $P_{3,b} = (2N_b - R_{3,b})/3$ and apply the mixed covariance bound. In the three-coordinate primitive indicator, the two characters of exact order three in w, with both B-characters trivial, have coefficients $-1/2$ each. Their combined contribution is $-R_{3,b}/2$. The trivial triple has coefficient one, and the remaining absolute coefficient mass is $M - 2$. Hence

$$D_{d,b} = c(N_b - R_{3,b}/2 + Z_b), \qquad |Z_b| \leq 8(M-2)\sqrt{t}.$$

The Hasse bound for N_b and the genus-two bound for $R_{3,b}$ give (4.11). Moreover,

$$\mathrm{Var}(N_b - R_{3,b}/2) \leq V_N + 4t + (C_0 + 32)W(A)t^{3/2}/A.$$

The centered L^2 triangle inequality now proves the stated full-order bound. The final form uses $V_N/t \to 1$ and $W(A) = t^{o(1)}$. □

These results isolate cancellation for one genuine coordinate of the w sieve. The full-order variance bounds retain the factor $W(A)^2$ through M; no main term for that variance is asserted.

5 Uniform relative traces and the exceptional trace groups

The two full-order conditions become equivalent when the linked elements are Frobenius conjugates. We first allow their difference to be any nonzero $c \in E$, obtaining a distribution theorem on the whole relative-trace field. We classify its two exceptional trace groups. Their intersections with K^* are both $\{1\}$, recovering the two exceptional trace-one parameters. The case $c = 1$ is a subfamily of D_d. We then describe the distribution within each trace-norm class and transfer the classification to every nonzero K-norm fiber.

5.1 A uniform four-point character estimate

For $c \in E$ put

$$Q_c = \#\{a \in C_d : a + a^{t^2} = c\}, \qquad M_c = \#\{a \in C_d : \mathrm{ord}(a) = d,\ a + a^{t^2} = c\}.$$

The notation M_c agrees with the projective-to-affine comparison in Supplementary Appendix S.C. In particular $M_1 = \Delta_t$ for Proposition S7.1.

Theorem 5.1 (Uniform nonzero relative-trace slices)**.** *Let $t = 2^s$, $s \geq 2$, and $c \in E^*$. The map*

$$a = \frac{c}{1+\gamma}, \qquad \tau = \gamma + \gamma^{-1}, \qquad w = \tau/c^2, \tag{5.1}$$

is a bijection from $\{\gamma \in V : w \in U\}$ onto the set counted by Q_c. It restricts to a bijection onto the set counted by M_c when $\mathrm{ord}(\gamma) = B$ and $\mathrm{ord}(w) = A$. For every nontrivial character pair (η, χ) of $U \times V$,

$$\left| \sum_{\substack{\gamma \in V \\ (\gamma+\gamma^{-1})/c^2 \in U}} \eta((\gamma+\gamma^{-1})/c^2)\chi(\gamma) \right| \leq 4\sqrt{t}. \tag{5.2}$$

Consequently, uniformly in $c \in E^*$,

$$|M_c - \theta(d)Q_c| \leq 4\theta(d)(W(d)-1)\sqrt{t}, \tag{5.3}$$

$$|Q_c - (t+1)| \leq 2\sqrt{t}, \tag{5.4}$$

$$|M_c - \theta(d)(t+1)| \leq \theta(d)(4W(d)-2)\sqrt{t}. \tag{5.5}$$

The zero slice contains no element of order d: $M_0 = 0$.

Proof. If $\gamma \in V$ and $w \in U$, then $\gamma \neq 1$ and

$$a^{t^2} = a + c, \qquad a^B = c^2/\tau = w^{-1}, \qquad a^{t^2-1} = \gamma.$$

Thus $a \in C_d$. Conversely an element counted by Q_c has $\gamma = (a+c)/a \in V$ and satisfies the same identities, so the maps are inverse. Since $\gcd(A, B) = 1$ and $\gcd(B, t^2-1) = 1$, the displayed powers recover the A-part and B-part of the order of a. Thus a has order d exactly when w and γ have orders A and B. If $c = 0$, then $a \in E$ and its order divides $\gcd(d, t^2-1) = A < d$, proving the final assertion.

Every $c \in E^*$ has a unique decomposition $c = c_0 u$ with $c_0 \in K^*$ and $u \in U$: take $c_0^2 = N_{E/K}(c)$ and $u = c/c_0$. Writing $w_0 = \tau/c_0^2$, we have $w = u^{-2}w_0$. The two summation sets are identical, and their character sums satisfy

$$S_c(\eta, \chi) = \eta(u^{-2})S_{c_0}(\eta, \chi), \qquad Q_c = Q_{c_0}.$$

The extra factor has absolute value one. It therefore suffices to establish the character estimate for $c \in K^*$, which we assume during the following geometric argument. The primitive indicator will still be applied to the original $w = u^{-2}w_0$; the full-order populations M_c need not equal M_{c_0}.

Over $\overline{K}$, compactify the parameter curve in $\mathbb{P}^1 \times \mathbb{P}^1$ as

$$Y_c: \quad (x+1)^2(y+1)^2 = c^4xy, \qquad F_Y(x, y) = (y^{-t}, x^t). \tag{5.6}$$

The displayed Frobenius is the inherited torus descent; it is not coordinatewise Frobenius on a split model. Indeed, $\iota(x, y) = (y^{-1}, x)$ preserves the compactification, commutes with coordinatewise t-power Frobenius and satisfies $\iota^4 = \mathrm{id}$. Thus $F_Y = \iota \circ \mathrm{Frob}_t$ gives the indicated descent over the quartic splitting field, with $F_Y^4 = \mathrm{Frob}_{t^4}$. Its open K-points have $x = \gamma$, $y = \gamma^t$, $\gamma \in V$, and $N_{E/K}(\tau) = c^4$. Hence they are precisely the summation set. The affine partial derivatives are c^4y, c^4x; their only common zero is off the curve. The boundary points are $P_0 = (0, 1)$, $P_\infty = (\infty, 1)$, $Q_0 = (1, 0)$, $Q_\infty = (1, \infty)$. Writing $u = y + 1$ at P_0 gives $(1+x)^2u^2 = c^4x(1+u)$, so the derivative with respect to x is $c^4 \neq 0$ and $v_{P_0}(x) = 2$. Reciprocal and coordinate symmetries give smoothness

at the other three points. Thus the bidegree-$(2,2)$ curve is smooth. It is geometrically connected: distinct smooth components would be disjoint, whereas any decomposition of the bidegree $(2,2)$ into nonzero effective bidegrees contains two components with positive intersection. It is therefore geometrically irreducible, and adjunction gives genus one. Frobenius cycles

$$P_0 \longmapsto Q_0 \longmapsto P_\infty \longmapsto Q_\infty \longmapsto P_0.$$

The boundary has no K-point, and Hasse–Weil proves (5.4).

To realize the character sum by the actual Lang covers, set

$$w = \frac{(x+1)^2}{c^2 x}, \qquad h = \frac{y^t}{x}.$$

The two split coordinates of the map to $\ker N_{E/K}$ are (w, w^{-1}). As in the Lang construction [4], the pullbacks from $\ker N_{E/K}$ and $\mathrm{Res}_{E/K}(\ker N_{L/E})$ are

$$z_U^A = w^{-1}, \qquad z_V^B = h.$$

Indeed their split Lang equations are respectively $z_U^{-t}/z_U = w$ and $z_1^{-t}/z_0 = x$, $z_0^t/z_1 = y$. Here $z_V = z_0$: substituting $z_1 = z_V^t/y$ into the first Lang equation on the second torus gives $z_V^B = y^t/x$. The covering Frobenius sends (z_U, z_V) to (wz_U, xz_V) over F_Y; the identities $w(F_Y) = w^{-t}$ and $h(F_Y) = x^B h$ verify compatibility. Thus its deck multipliers at a K-point are w and γ, and the character trace is $\eta(w)\chi(\gamma)$, with the inverse convention merely inverting both characters.

Multiplication by the nonzero constant c^{-2} changes no boundary valuation. At Q_0, the local equation gives $v(y) = 2$ and $v(x+1) = 1$; inversion and coordinate interchange give the other rows. The valuations are

	w	h
P_0	-2	-2
P_∞	-2	2
Q_0	2	$2t$
Q_∞	2	$-2t$

(5.7)

For $0 \le e_0 < A$, $0 \le e_1 < B$, an ABth-power relation $(w^{-1})^{Be_0} h^{Ae_1} \in \overline{K}(Y_c)^{\times AB}$ would have valuation $2Be_0 - 2Ae_1$ at P_0. Reduction modulo A and B forces $e_0 = e_1 = 0$. The product cover is therefore geometrically connected with full group $U \times V$. The inertia vectors for the Kummer functions (w^{-1}, h) are $(2,-2), (2,2), (-2,2t), (-2,-2t)$, with coordinates modulo A, B. Since $\gcd(A,2) = \gcd(B,2t) = \gcd(A,B) = 1$, each vector generates $U \times V$. Every nontrivial character pair is therefore ramified at all four boundary points. The ramification is tame, with local conductor one. These four geometric points form one closed point of degree four, so the conductor has degree four. The associated rank-one finite-monodromy sheaf is geometrically nontrivial and pure of weight zero. Its H_c^0, H_c^2 vanish; Grothendieck–Ogg–Shafarevich gives $\dim H_c^1 = 2g - 2 + 4 = 4$. The classical Artin-character estimate [5, §4, Lemma 1 in the author's version] now gives (5.2). Thus the Lang construction identifies the character trace, while the curve estimate supplies its bound.

The phase identity now gives the same character bound for every $c \in E^*$. The product of the two primitive indicators, applied to the original pair (w, γ), has constant coefficient $\theta(A)\theta(B) = \theta(d)$ and total absolute coefficient mass $\theta(d)W(d)$. The nonconstant terms give (5.3). Adding the Hasse error of the constant term gives (5.5). The identity $Q_c = Q_{c_0}$ transfers the unfiltered estimate as well. All constants are therefore uniform on E^*. □

The classical reciprocal-trace substitution [17, §4], combined with the smoothness, connectedness and ramification checks above, makes the curve-character theorem uniform on these exact-order

slices. Only $c = 1$ makes $a + c = 1 + a$; the other slices describe relative-trace distributions. The direct counting predecessors and the already known $s = 2$ positivity case [25, Theorem 1.6] are compared in Appendix E.

5.2 The exact unfiltered center

The center Q_c can be written as a classical binary Kloosterman sum. This separates the fluctuation of the whole slice from the error caused by the order restrictions. Put $\psi(z) = (-1)^{\mathrm{Tr}_{K/\mathbb{F}_2}(z)}$ and $\mathrm{Kl}_t(\lambda) = \sum_{z \in K^*} \psi(z + \lambda/z)$.

Lemma 5.2 (Unfiltered trace slices)**.** *For every $c \in E^*$,*

$$Q_c = t + 1 + \mathrm{Kl}_t(N_{E/K}(c)^{-2}). \tag{5.8}$$

Proof. First assume $c \in K^*$. For $b \in U$, the polynomial $X^2 + cX + b$ gives two roots with relative trace c exactly when it is irreducible over E. The binary quadratic criterion therefore gives

$$Q_c = A - \sum_{b \in U} \psi((b + b^{-1})/c^2).$$

For every $r \in K$, the total number of roots of $X^2 + rX + 1$ in the two groups K^* and U is two. For $r \neq 0$ the polynomial splits in one of the two tori; at $r = 0$, its root 1 is counted once in each. Hence, for $\alpha \neq 0$, the two torus character sums add to $2\sum_{r \in K} \psi(\alpha r) = 0$. The sum over K^* equals $\mathrm{Kl}_t(\alpha^2)$ after $z \mapsto \alpha z$. Taking $\alpha = c^{-2}$ gives $Q_c = t + 1 + \mathrm{Kl}_t(c^{-4})$. For general $c = hu$, the preceding proof gives $Q_c = Q_h$ and $h^{-4} = N_{E/K}(c)^{-2}$, proving the stated identity. □

5.3 A prime sieve and complete trace classification

The following is the usual prime-sieve inequality for freeness [22, eq. (3.2)]; see also [7, 13]. Here it is applied after the four-point estimate. We include the calculation to specify its constant.

Lemma 5.3 (Prime sieve on a nonzero trace slice)**.** *Let $e \mid \mathrm{rad}(d)$ be a squarefree core and let $p_1, \dots, p_r$ be the primes dividing d but not e. Put*

$$\delta = 1 - \sum_{i=1}^{r} \frac{1}{p_i} > 0, \qquad \Lambda = 2 + \frac{r - 1}{\delta}.$$

For $r = 0$ these definitions give $\delta = \Lambda = 1$. For every $c \in E^*$,

$$M_c \geq \theta(e)\delta \left[t + 1 - \big(4W(e)\Lambda - 2\big)\sqrt{t} \right]. \tag{5.9}$$

Proof. Identify $U \times V$ with a cyclic group of order d and let $N_e(c)$ count those (w, γ) on Y_c avoiding its pth-power subgroup for every $p \mid e$. The primitive indicator with the primes of e gives

$$|N_e(c) - \theta(e)A| \leq \theta(e)(4W(e) - 2)\sqrt{t}.$$

For $p \nmid e$, the indicator for ep minus $(1 - p^{-1})$ times the indicator for e contains only nontrivial character pairs, of total absolute coefficient mass $\theta(e)(1 - p^{-1})W(e)$. Thus

$$|N_{ep}(c) - (1 - p^{-1})N_e(c)| \leq 4\theta(e)(1 - p^{-1})W(e)\sqrt{t}.$$

The pointwise union bound for the omitted-prime failures gives

$$M_c \geq \sum_{i=1}^{r} N_{ep_i}(c) - (r-1)N_e(c).$$

Writing the right side as $\delta N_e(c)$ plus the r differences just estimated, and using $\sum_i (1-p_i^{-1}) = r-1+\delta$, proves the bound. For $r = 0$ it reduces to the first displayed estimate. □

To pass from a representative trace $h \in K^*$ to its whole trace-norm class, we use the reciprocal-trace supports. For $h \in K^*$ define

$$T_h = \{(\gamma + \gamma^{-1})/h^2 \in U : \gamma \in V,\ \mathrm{ord}(\gamma) = B\}, \qquad n_h = 2|T_h|.$$

The factor two records the pair γ, γ^{-1} over each reciprocal trace. The class hU consists of exactly those $c \in E^*$ with $N_{E/K}(c) = h^2$. They satisfy $T_h = T_h^{-1}$: raising a primitive γ to the tth power preserves its order, fixes h, and sends $(\gamma + \gamma^{-1})/h^2$ to its inverse in U. This Frobenius symmetry is essential to the following reduction. The underlying Chinese remainder description is classical in unitary Cayley graphs; see [28, §1]. We give the elementary argument needed here.

Lemma 5.3a (Inverse supports with at most two prime divisors)**.** *Let G be cyclic of odd order $n > 1$, let $T = T^{-1} \subseteq G$, and put $\rho = \mathrm{rad}(n)$. Let H have order n/ρ, let S be the image of T in G/H, and define*

$$F_T(u) = 2\#\{z \in T : \mathrm{ord}(zu^{-2}) = n\}.$$

If $\rho = p$ is prime, the zero set of F_T is G when $T = \varnothing$, is G^p when $S = \{1\}$, and is empty otherwise.

Suppose $\rho = pq$ for distinct primes and $F_T(1) > 0$. Then $F_T(u) > 0$ for every $u \in G$, unless $S = \{v, v^{-1}\}$ for a generator v of G/H. In that exceptional case write $v = (a, b)$ under $G/H \cong C_p \times C_q$. The zeros are exactly the u with

$$\overline{u}^{\,2} \in \{(a, b^{-1}), (a^{-1}, b)\},$$

and there are $2n/(pq)$ of them. In particular $|S| \geq 3$ excludes this case; the sufficient condition $|T| > 2n/\rho$ also does so.

Finally, if $\rho = pq$, $S \subseteq \{1\} \times C_q$ and $|S| \geq 2$, the zero set is exactly G^p.

Proof. An element generates G exactly when its image has nonidentity coordinate at every prime dividing ρ. Hence

$$F_T(u) = 0 \quad \Longleftrightarrow \quad T \subseteq \bigcup_{p|\rho} u^2 G^p.$$

For $\rho = p$, a nonempty S can be covered by one coordinate value only when it is a singleton. Inversion then forces that value to be 1, since p is odd. Squaring is bijective, which proves the first assertion.

For $\rho = pq$ and $F_T(1) > 0$, choose $v = (a, b) \in S$ with $a, b \neq 1$. Inversion puts v^{-1} in S, and oddness gives $a \neq a^{-1}$, $b \neq b^{-1}$. A zero must cover both points by its row and column. Its squared image must therefore be one of the two mixed corners (a, b^{-1}), (a^{-1}, b). The zero set is inversion invariant, so if one mixed corner occurs then both occur. The intersection of their row–column unions is precisely $\{v, v^{-1}\}$. Thus S must be that pair. Conversely that pair has exactly the two stated zero corners. Each lifts to n/ρ elements, proving the formula and the cardinality consequences.

For the final assertion, the first coordinate 1 covers all of S. Any other first coordinate would require its column to cover two distinct points of S, which is impossible. □

By Theorem 5.1, $M_{hu} = F_{T_h}(u)$. Thus, when A has two distinct prime divisors, a primitive support point z_0 and one additional point whose image is outside $\{\overline{z}_0, \overline{z}_0^{-1}\}$ certify positivity throughout hU. The additional condition cannot be dropped: in $G = \langle g \rangle$ of order 15, $T = \{g, g^{-1}\}$ has $F_T(1) = 4$ but $F_T(g^2) = F_T(g^{13}) = 0$. This is a cyclic-group reduction, not an additional geometric estimate. Neither its proof nor the support certificates below uses the classification to be proved.

Theorem 5.4 (Complete nonzero-relative-trace classification)**.** *For $s \geq 2$ and $c \in E^* = \mathbb{F}_{2^{2s}}^*$,*

$$M_c = 0 \quad \Longleftrightarrow \quad (s = 3 \text{ and } c^9 = 1) \quad \text{or} \quad (s = 5 \text{ and } c^{11} = 1). \tag{5.10}$$

Thus the exceptional sets are the subgroup of order 9 at $s = 3$ and the subgroup of order 11 at $s = 5$. The analytic estimate and prime sieve give positivity in every nonzero slice for $s \geq 10$, and also for $s = 8$. On restricting to K^*, each exceptional subgroup meets K^* only at 1. In particular the trace-one classification is

$$\Delta_{2^s} = M_1 > 0 \quad \Longleftrightarrow \quad s \notin \{3, 5\} \qquad (s \geq 2). \tag{5.11}$$

Proof. For $s \geq 33$, the independent arithmetic Lemma B.1 gives $\sqrt{t} \geq 4W(d)$. Theorem 5.1 then proves $M_c > 0$ for every $c \neq 0$. For $s = 8$ and $10 \leq s \leq 32$, Appendix C supplies a core $e \in \{1, 3\}$ and an integer $C_s \geq 4W(e)\Lambda - 2$ with $(t+1)^2 > tC_s^2$. Lemma 5.3 proves positivity in all these slices.

Let Φ_j denote the reduction modulo two of the jth cyclotomic polynomial and put $H_j(X) = X^{\varphi(j)}\Phi_j(X + X^{-1})$. This classical Q-transform [24, §1] records the exact order of a reciprocal trace. Since B is odd, Φ_B has simple roots, all primitive in V. In particular

$$\Delta_t = \deg \gcd(\Phi_B, H_A) = \sum_{e \mid \mathrm{rad}(B)} \mu(e) \deg \gcd(X^{B/e} + 1, H_A). \tag{5.12}$$

Appendix C retains the exact small trace-one counts. For the exceptional supports, exact binary polynomial divisions give

$$\begin{aligned} \gcd(H_j, \Phi_{65}) &= 1 \qquad (j = 1, 3, 9), \\ \gcd(H_j, \Phi_{1025}) &= 1 \qquad (j = 1, 3, 33), \\ H_{11} &\mid \Phi_{1025}. \end{aligned}$$

Thus $T_1 = \varnothing$ at $t = 8$, since $1, 3, 9$ are all possible reciprocal-trace orders in U. At $t = 32$, the divisibility $H_{11} \mid \Phi_{1025}$ makes its twenty roots distinct, because 1025 is odd. Each nonzero reciprocal trace τ has exactly the two distinct preimages γ, γ^{-1}, the roots of $X^2 + \tau X + 1$. These roots therefore give all ten order-11 reciprocal traces, while the other possible orders $1, 3, 33$ have no primitive preimage. Hence $T_1 = C_{11} \setminus \{1\}$. The inverse-support lemma now gives exactly the empty group $C_9 = U$ in the first case, and $U^3 = C_{11}$ in the second. For the positive-population refinement below, the same divisions give $H_1 = \Phi_3$, $H_3 = \Phi_5$ and

$$\gcd(H_{33}, X^{1025} + 1) = \gcd(H_{33}, X^{205} + 1), \qquad \deg \gcd(H_{33}, X^{205} + 1) = 20,$$

with $\gcd(H_{33}, X^{41} + 1) = \gcd(H_{33}, X^5 + 1) = 1$. These identities also identify the order drop in the latter support obstruction.

For the remaining finite range $s = 2, \ldots, 7, 9$, every $A = t + 1$ has at most two distinct prime divisors. Appendix C supplies 755 primitive support witnesses, one for every $h \in K^*$ except $(s, h) = (3, 1), (5, 1)$. They prove $M_h > 0$ directly through Theorem 5.1. In the 731 classes where A has two distinct prime divisors, a second witness has image outside the inverse pair of the first, so $|\overline{T_h}| \geq 3$. The inverse-support lemma gives $M_{hu} > 0$ for every $u \in U$. The independent verifier

checks both kinds of witnesses in degree-$4s$ binary polynomial fields, without an all-E coverage mask. Unique decomposition $c = hu$ proves every remaining slice positive. Together with the analytic range and the two exact support obstructions, this proves (5.10). Restriction to K^*, which meets each exceptional group only at 1, proves (5.11). The earlier 3319 full-trace witness families remain a separate check, rather than a premise of this reduction. □

Every trace in either exceptional group has norm 1. Lemma 5.2 and $Q_c = Q_1$ therefore give $Q_c = 4$ at $s = 3$ and $Q_c = 44$ at $s = 5$ on these groups. The unfiltered slices are nonempty; the exact-order filter causes the vanishing.

5.4 Distribution within a fixed trace-norm class

The preceding classification concerns zeros. A cyclic convolution also describes the positive populations and their fluctuation. Retain T_h and n_h from the inverse-support reduction above.

Proposition 5.5 (Cyclic convolution and a variance bound)**.** *For $h \in K^*$ and $u \in U$,*

$$M_{hu} = 2\#\{z \in T_h : \operatorname{ord}(zu^{-2}) = A\}. \tag{5.13}$$

The mean $\mu_h = A^{-1}\sum_{u\in U} M_{hu}$ and variance satisfy

$$\mu_h = \theta(A)n_h, \tag{5.14}$$

$$\frac{1}{A}\sum_{u\in U} |M_{hu} - \mu_h|^2 \le 16\theta(A)(1-\theta(A))\theta(B)^2W(B)^2t. \tag{5.15}$$

Moreover $M_{huv} = M_{hu}$ for every v in the subgroup of U of order $A/\operatorname{rad}(A)$. If $n_h > 0$, the number Z_h of empty slices in hU satisfies

$$\frac{Z_h}{A} \le \frac{16\theta(A)(1-\theta(A))\theta(B)^2W(B)^2t}{\mu_h^2}.$$

Proof. In Theorem 5.1, the admissible primitive γ have $z = (\gamma + \gamma^{-1})/h^2 \in U$. Each z has exactly two preimages, since the quadratic $X^2 + h^2zX + 1$ has distinct roots γ, γ^{-1}. The remaining order condition is that zu^{-2} generate U, proving (5.13). As $u \mapsto u^2$ permutes U, each z contributes to precisely $\varphi(A)$ traces. This gives the mean.

For a character η of U put

$$C_A(\eta) = \sum_{\operatorname{ord}(w)=A} \overline{\eta(w)}, \qquad S_h(\eta) = 2\sum_{z\in T_h} \eta(z).$$

The classical Ramanujan-sum identity gives $C_A(\eta) = \mu(j)\varphi(A)/\varphi(j)$ when $j = \operatorname{ord}(\eta)$. Fourier inversion on U yields

$$M_{hu} = \frac{1}{A}\sum_{\eta\in\widehat{U}} C_A(\eta)S_h(\eta)\eta(u^{-2}).$$

For $\eta \neq 1$, expand only the primitive indicator on V. All resulting character pairs are nontrivial, so Theorem 5.1 gives $|S_h(\eta)| \le 4\theta(B)W(B)\sqrt{t}$. Orthogonality, again using the permutation $u \mapsto u^2$, therefore gives the exact identity

$$\frac{1}{A}\sum_{u} |M_{hu} - \mu_h|^2 = \frac{1}{A^2}\sum_{\eta\neq 1} |C_A(\eta)|^2|S_h(\eta)|^2.$$

Parseval for the primitive indicator says $\sum_\eta |C_A(\eta)|^2 = A\varphi(A)$. Subtracting its trivial-character term $\varphi(A)^2$ proves (5.15). Only characters of squarefree order contribute. They are trivial on the subgroup of order $A/\operatorname{rad}(A)$, proving the claimed period. Each zero contributes μ_h^2 to the squared deviation, which proves the final bound. □

The Fourier identities are standard. Their use here retains the four-point geometric estimate and removes $W(A)$ from the variance bound. The equality $Q_{hu} = Q_h$ thus coexists with a nonconstant distribution of the exact-order counts.

Corollary 5.6 (The exceptional norm-one profiles)**.** *For $t = 8$, $T_1 = \varnothing$ and $M_c = 0$ for every $c \in U = C_9$. For $t = 32$, $T_1 = C_{11} \setminus \{1\} \subset U = C_{33}$ and*

$$M_c = \begin{cases} 0, & c \in C_{11}, \\ 20, & c \in C_3 \setminus \{1\}, \\ 18, & c \in U \setminus (C_{11} \cup C_3). \end{cases} \tag{5.16}$$

The three populations occur on 11, 2, and 20 traces, respectively. Throughout U, the unfiltered counts are 4 at $t = 8$ and 44 at $t = 32$.

Proof. For $t = 8$ the reciprocal-trace exclusions in the proof of Theorem 5.4 already show $T_1 = \varnothing$. For $t = 32$, let $H_j(X) = X^{\varphi(j)}\Phi_j(X + X^{-1})$ as above. Exact polynomial division gives the following complete table for $\gamma \in V$ and $\tau = \gamma + \gamma^{-1} \in U$:

$\operatorname{ord}(\tau)$	1	3	11	33
$\#\gamma$	0	4	20	20
$\operatorname{ord}(\gamma)$	–	5	1025	205.

The $j = 1, 3, 33$ entries are the polynomial identities already used above. For the remaining entry, H_{11} has degree 20 and divides Φ_{1025} over $\mathbb{F}_2$; the accompanying independent verifier checks the division. Thus all ten order-11 reciprocal traces, and no others in U, arise from primitive γ. This proves $T_1 = C_{11} \setminus \{1\}$.

More generally let $U = C_\ell \times C_p$ for distinct odd primes ℓ, p, and suppose $T_h = C_p \setminus \{1\}$. If $u \in C_p$, the C_ℓ component of zu^{-2} is trivial, so the convolution is zero. If $u \in C_\ell \setminus \{1\}$, all $p - 1$ choices of z give full order. In the remaining case exactly one choice, $z = u_p^2$, has trivial C_p component; the other $p - 2$ choices give full order. Formula (5.13), with $(\ell, p) = (3, 11)$, gives (5.16). The unfiltered counts follow from the table and the $t = 8$ order-3 reciprocal traces. □

5.5 Transfer to arbitrary nonzero norm fibers

The subgroup count has a direct interpretation on every coset of C_d. This transfer uses the coprime factorization $L^* = K^* \times C_d$. Generator counts in cyclic cosets are classical; see, for example, [11, Lemma 1]. The transfer below is a consequence of the preceding theorems, rather than a separate character-sum method.

Corollary 5.7 (Prescribed norm and relative trace)**.** *Let $\nu \in K^*$ have order e, and let $\lambda \in K^*$ be the unique element with $\lambda^4 = \nu$. The greatest order attained on $N_{L/K}^{-1}(\nu)$ is ed. Put*

$$M_{\nu,c} = \#\{a \in L^* : N_{L/K}(a) = \nu,\ a + a^{t^2} = c,\ \operatorname{ord}(a) = ed\}.$$

Then $M_{\nu,0} = 0$ and, for $c \in E^*$,

$$M_{\nu,c} = M_{c/\lambda}, \qquad |M_{\nu,c} - \theta(d)(t+1)| \le \theta(d)(4W(d) - 2)\sqrt{t}. \tag{5.17}$$

In particular,

$$M_{\nu,c} = 0 \quad \Longleftrightarrow \quad (s = 3 \text{ and } c^{36} = \nu^9) \quad \text{or} \quad (s = 5 \text{ and } c^{44} = \nu^{11}). \tag{5.18}$$

The unfiltered population with the same norm and trace is

$$Q_{\nu,c} = t + 1 + \mathrm{Kl}_t(\nu/N_{E/K}(c)^2).$$

If ν is primitive in K^*, these are counts of primitive elements of the ambient field L.

Proof. The integers $t-1$ and d are coprime. Every element of the norm fiber has a unique expression $a = \lambda x$, $x \in C_d$. The fourth-power map permutes K^*, so $\mathrm{ord}(\lambda) = e$ and $\mathrm{ord}(a) = e\,\mathrm{ord}(x)$. Its relative trace is $\lambda(x + x^{t^2})$. The bijection proves the count identity, the maximum-order assertion and the bound. The exceptional sets are therefore λC_9 at $s = 3$ and λC_{11} at $s = 5$. Taking fourth powers gives the displayed equations, with equivalence because fourth powers also permute E^*. Finally $Q_{\nu,c} = Q_{c/\lambda}$ and $N_{E/K}(c/\lambda)^{-2} = \nu/N_{E/K}(c)^2$. Lemma 5.2 gives the final formula. When $e = t - 1$, the maximum order ed equals $t^4 - 1$. □

5.6 Consequences for adjacent pairs and affine groups

The trace-one specialization and the two off-slice certificates give an alternative proof of the all-parameter existence assertion in Theorem 4.4. The stronger Theorem 4.5 proves existence at each primitive inner norm independently of that route. At the exceptional parameters $s = 3, 5$, the full-family populations are $D_d = 24, 240$. Appendix B retains the wider certificate range $2 \le s \le 32$ as additional exact checks.

Every slope counted by D_d lies in no proper subfield of L, because $d > t^2 - 1$. Thus $\mathrm{ord}_d(2) = 4s$, and Corollary 2.5 makes each of these slopes, with any nonzero translation, a realization of $L^+ \rtimes C_d$. For elements counted by M_c, the enlarged trace range distinguishes two normality types over K. Put $T(a) = a^t$; then $(T+1)^2 a = c$ and $(T+1)^3 a = c + c^t$. For $c \in K^*$ the K-order is $(X+1)^3$, so these are 1-normal elements. For $c \in E \setminus K$ it is $(X+1)^4$, so the elements are normal over K; the latter criterion is classical for binary quartic extensions [11, §3]. The 1-normal terminology follows [21]. The same estimate thus treats both normality types through their relative traces. Their $O(t)$ slice populations and the $O(t^2)$ adjacent-pair populations describe different parts of the norm geometry.

6 General index bounds and the norm obstruction

We return to $q = 2^n$, $d \mid q-1$ and $k = (q-1)/d$. The quartic result treats the special index $q^{1/4} - 1$. For comparison, this section gives an elementary sufficient range for all admissible indices and a structural obstruction just beyond the quarter-power scale.

6.1 Jacobi bounds and fixed indices

Proposition 6.1 (A positive lower bound for realizing slopes)**.** *Let $q = 2^n$, let $d > 1$ divide $q - 1$, and set $k = (q-1)/d$. Write $\omega(d)$ for the number of distinct prime divisors of d, and put*

$$b_k = \frac{(k-1)(k-2)}{k^2}.$$

For every divisor e of $q - 1$, with $\ell = (q-1)/e$, one has

$$\left| T_e - \frac{q + 1 - 3\ell}{\ell^2} \right| \le b_\ell \sqrt{q}. \tag{6.1}$$

Consequently,

$$|\mathcal{R}_d| \geq \frac{q}{k^2}\left(1 - \sum_{p\mid d}\frac{1}{p^2}\right) - \frac{3k-1}{k^2} - (\omega(d)+b_k)\sqrt{q}, \tag{6.2}$$

where the sum runs over the distinct prime divisors of d. In particular,

$$|\mathcal{R}_d| > \frac{3q}{4k^2} - \frac{3k-1}{k^2} - (\omega(d)+b_k)\sqrt{q}. \tag{6.3}$$

Proof. The input is the classical Jacobi-sum calculation [3, Theorems 1–2]; we include its character-sum proof. Choose a character χ of order ℓ, and extend all its powers, including the trivial character, by zero at zero. The subgroup indicator gives

$$T_e = \frac{1}{\ell^2}\sum_{i,j=0}^{\ell-1} J(\chi^i,\chi^j), \qquad J(\eta,\theta) = \sum_{x\in\mathbb{F}_q}\eta(x)\theta(1+x).$$

The term $i=j=0$ is $q-2$. Each of the $2(\ell-1)$ terms with exactly one nontrivial character is -1. The $\ell-1$ terms with both characters nontrivial and product trivial are also -1: the substitution $u = x/(1+x)$ identifies their sum with the sum of a nontrivial character over $\mathbb{F}_q^\times \setminus \{1\}$.

For completeness, fix a nontrivial additive character ψ and set $G(\eta) = \sum_{x\neq 0}\eta(x)\psi(x)$. For nontrivial η, substituting $x = ty$ into its squared modulus gives

$$|G(\eta)|^2 = \sum_{t\neq 0}\eta(t)\sum_{y\neq 0}\psi((t+1)y) = (q-1) - \sum_{t\neq 0,1}\eta(t) = q.$$

If η, θ, and $\eta\theta$ are all nontrivial, grouping $G(\eta)G(\theta)$ by $z = x+y$ gives

$$G(\eta)G(\theta) = J(\eta,\theta)G(\eta\theta).$$

Here the $z=0$ contribution vanishes by multiplicative-character orthogonality; for $z\neq 0$ put $x = zu$, $y = z(1+u)$. Thus $|J(\eta,\theta)| = \sqrt{q}$. There are $(\ell-1)(\ell-2)$ remaining terms. Their triangle bound, together with the terms evaluated above, proves (6.1), also when $\ell = 1$.

A pair counted by T_d that does not generate C_d lies in at least one maximal subgroup $C_{d/p}$, for a prime $p \mid d$. Therefore

$$|\mathcal{R}_d| \geq T_d - \sum_{p\mid d} T_{d/p}.$$

Applying (6.1) to the first term gives $T_d \geq (q+1-3k)/k^2 - b_k\sqrt{q}$. For the subgroup of index kp, the same bound gives

$$T_{d/p} \leq \frac{q+1-3kp}{k^2p^2} + b_{kp}\sqrt{q} \leq \frac{q}{k^2p^2} + \sqrt{q}.$$

This proves (6.2). All the primes dividing d are odd. The elementary telescoping bound

$$\sum_{p\mid d}\frac{1}{p^2} \leq \sum_{j=1}^{\infty}\frac{1}{(2j+1)^2} < \frac{1}{4}\sum_{j=1}^{\infty}\frac{1}{j(j+1)} = \frac{1}{4}$$

proves (6.3). □

Theorem 6.2 (Realization at any fixed odd index)**.** *Let $k > 1$ be odd. For every sufficiently large n satisfying $k \mid 2^n - 1$, put $q = 2^n$ and $d = (q-1)/k$. Then*

$$H_d = \mathbb{F}_q^+ \rtimes C_d$$

is a primitive subgroup of index k in $\mathrm{AGL}(1,q)$ *and is generated by f and $F(f)$ for a scalar affine complete mapping. An explicit sufficient condition is $d > \sqrt{q} - 1$ and*

$$L_{q,k,d} := \frac{3q}{4k^2} - \frac{3k-1}{k^2} - (\omega(d) + b_k)\sqrt{q} > 0. \tag{6.4}$$

Under these conditions the number of realizing maps in the specified field coordinates is $(q-1)|\mathcal{R}_d| > (q-1)L_{q,k,d}$. In particular, each fixed odd index $k > 1$ gives an infinite family of realizable proper primitive affine groups.

Proof. The degree $s = \mathrm{ord}_d(2)$ of the field generated by C_d divides n. If $s < n$, then $s \le n/2$ and $d \mid 2^s - 1 \le \sqrt{q} - 1$, contrary to the assumed inequality. Hence $s = n$. Proposition 6.1 and (6.4) imply $\mathcal{R}_d \neq \varnothing$. Corollaries 2.5 and 2.4 give the realization, primitivity, and exact count. The index is k because the translation subgroup is full and the complement has order $(q-1)/k$.

To establish the infinite-family assertion without an arithmetic assumption on d, note that $3^{\omega(d)} \le d < 2^n$. Since $3 > 2^{3/2}$, this yields $\omega(d) < 2n/3$. Also $0 \le b_k \le 1$. For fixed k, the positive term $3 \cdot 2^n/(4k^2)$ in (6.4) eventually exceeds the constant term and $(2n/3+1)2^{n/2}$. Likewise $(2^n-1)/k > 2^{n/2} - 1$ eventually. There are infinitely many admissible n, namely multiples of $\mathrm{ord}_k(2)$. The sufficient conditions therefore hold for every sufficiently large admissible dimension. □

6.2 A quantitative divisor sieve and growing indices

Theorem 6.3 (Quantitative slope counts)**.** *Let $q = 2^n$, $d > 1$ divide $q - 1$, and $k = (q-1)/d$. Set*

$$\theta_j(d) = \prod_{p \mid d}(1 - p^{-j}) \quad (j = 1, 2), \qquad S_{k,d} = \sum_{h \mid \mathrm{rad}(d)} b_{kh}, \qquad b_\ell = \frac{(\ell-1)(\ell-2)}{\ell^2}.$$

Here $\mathrm{rad}(d)$ *is the product of its distinct prime divisors. Then*

$$|\mathcal{R}_d| = \frac{q+1}{k^2}\theta_2(d) - \frac{3}{k}\theta_1(d) + E_{q,k,d}, \qquad |E_{q,k,d}| \le S_{k,d}\sqrt{q} \le 2^{\omega(d)}\sqrt{q}. \tag{6.5}$$

The sharper error coefficient is

$$S_{k,d} = 2^{\omega(d)} - \frac{3}{k}\prod_{p \mid d}(1 + p^{-1}) + \frac{2}{k^2}\prod_{p \mid d}(1 + p^{-2}). \tag{6.6}$$

For fixed k, as n tends to infinity through admissible dimensions,

$$|\mathcal{R}_d| = \frac{q}{k^2}\theta_2(d) + O_k(q^{1/2+o(1)}), \qquad \frac{|\mathcal{R}_d|}{T_d} = \theta_2(d) + O_k(q^{-1/2+o(1)}). \tag{6.7}$$

In particular $\theta_2(d) > 3/4$: asymptotically at least three quarters of the slopes satisfying the two subgroup-membership conditions generate the specified complement jointly.

Proof. Write $T_{d/h} = (q + 1 - 3kh)/(kh)^2 + \varepsilon_h$ in (2.1). Proposition 6.1 gives $|\varepsilon_h| \le b_{kh}\sqrt{q}$. Only squarefree h contribute. The identities

$$\sum_{h \mid d} \frac{\mu(h)}{h^j} = \theta_j(d), \qquad \sum_{h \mid \mathrm{rad}(d)} h^{-j} = \prod_{p \mid d}(1 + p^{-j})$$

give (6.5) and, since $b_\ell = 1 - 3/\ell + 2/\ell^2$, (6.6). All these coefficients are nonnegative and at most one for integer $\ell \ge 1$.

For each $\epsilon > 0$, choose a fixed cutoff beyond which $2 \le p^\epsilon$. The bounded contribution of smaller primes can be absorbed into a constant, so $2^{\omega(d)} \le C_\epsilon d^\epsilon$. Consequently $2^{\omega(d)} = q^{o(1)}$. The first asymptotic follows from (6.5); the omitted terms are bounded for fixed k. Also $T_d = q/k^2 + O_k(\sqrt{q})$ by (6.1), which gives the ratio assertion. Finally $\prod(1 - x_i) \ge 1 - \sum x_i$ for $0 \le x_i \le 1$, and the odd-prime bound in Proposition 6.1 is strictly less than $1/4$. Thus $\theta_2(d) > 3/4$. □

Corollary 6.4 (A uniform range of growing indices)**.** *Let $n \ge 16$, $q = 2^n$, and $k > 1$ divide $q - 1$. If*

$$k \le \frac{q^{1/4}}{\sqrt{n}}, \tag{6.8}$$

then $H_{(q-1)/k}$ is a realizable proper primitive affine companion group.

Proof. Put $d = (q-1)/k$. Since $k < \sqrt{q} + 1$, we have $d > \sqrt{q} - 1$, so the field-degree argument in Theorem 6.2 gives $\mathrm{ord}_d(2) = n$. The index is odd and at least three; hence $(3k-1)/k^2 < 1$. Using $\omega(d) < 2n/3$ and $b_k \le 1$ gives

$$L_{q,k,d} > \left(\frac{n}{12} - 1\right)\sqrt{q} - 1 > 0 \qquad (n \ge 16).$$

Theorem 6.2 proves the assertion. □

6.3 Subfield descent and the quarter-power barrier

Theorem 6.5 (An infinite family of primitive nonrealizable groups)**.** *Let $r \ge 2$, $q = 2^{2r}$, and $d = 2^r + 1$. The scalar affine group*

$$H_d = \mathbb{F}_q^+ \rtimes C_d \le \mathrm{AGL}(1, q)$$

is primitive of degree q, but no affine complete mapping f on $\mathbb{F}_2^{2r}$ satisfies $\langle f, F(f)\rangle = H_d$. The same nonrealizability holds for every $\mathrm{GL}(2r, 2)$-conjugate of this subgroup. More generally, if an odd integer $d > 3$ divides $2^r + 1$, then $\mathcal{R}_d = \varnothing$ in every characteristic-two field containing C_d.

Proof. Since $2^r \equiv -1 \pmod{d}$, the order of 2 modulo d divides $2r$. It cannot be at most r, as $0 < 2^j - 1 < d$ for $1 \le j \le r$. Every proper divisor of $2r$ is at most r, so $\mathrm{ord}_d(2) = 2r$. Consequently the multiplicative action of C_d on $\mathbb{F}_q$ is irreducible over $\mathbb{F}_2$, and its affine extension by all translations is primitive.

More generally, suppose $d > 3$ divides $2^r + 1$ and $a, 1 + a \in C_d$. Because $a^d = 1$ and $(1+a)^d = 1$, we have $a^{2^r} = a^{-1}$ and

$$1 = (1+a)^{2^r+1} = (1 + a^{-1})(1 + a) = a + a^{-1}.$$

Multiplying by a yields $a^2 + a + 1 = 0$. Thus a and $1 + a = a^2$ both have order 3, and $\langle a, 1 + a\rangle = C_3$. Since $d > 3$, no a belongs to $\mathcal{R}_d$. Corollary 2.5 proves that no scalar affine companion pair generates H_d. Moreover, every affine permutation belonging to the specified group H_d has precisely the scalar form $x \mapsto ax + c$, so this excludes arbitrary affine complete mappings whose generated group equals H_d.

Finally, for an invertible $\mathbb{F}_2$-linear change of coordinates S, one has $F(SfS^{-1}) = SF(f)S^{-1}$. Hence an affine companion realization of a conjugate subgroup would conjugate back to one of H_d, which is impossible. □

The theorem gives an obstruction beyond irreducibility and primitivity; it does not claim that all nonrealizable primitive affine groups have a complement of this form.

Theorem 6.6 (Norm descent into a proper subfield)**.** *Let $1 \le s < r$, $s \mid r$, $u \mid 2^s - 1$, and work in $L = \mathbb{F}_{2^{2r}}$. Put*

$$d = u(2^r + 1), \qquad g = \begin{cases} u, & r/s \text{ even}, \\ u(2^s + 1), & r/s \text{ odd}. \end{cases}$$

The two membership conditions descend exactly:

$$\{a \in C_d : 1 + a \in C_d\} = \{a \in C_g \subset \mathbb{F}_{2^{2s}}^{\times} : 1 + a \in C_g\}. \tag{6.9}$$

Thus $T_d = T_g$, $\mathcal{R}_d = \varnothing$, and $H_d = L^+ \rtimes C_d$ is primitive but cannot be generated by an affine complete mapping and its prescribed companion. The nonrealizability also holds for every linear conjugate of H_d.

Proof. Set $Q = 2^r$, $K = \mathbb{F}_Q$, and $E = \mathbb{F}_{2^s} \subset K$. The norm $N_{L/K}(x) = x^{Q+1}$ has kernel C_{Q+1} and maps C_d onto $C_u \subset E^{\times}$: its image has order $d/\gcd(d, Q+1) = u$, since $\gcd(u, Q+1) = 1$. If $a, 1 + a \in C_d$, put $U = N(a)$ and $W = N(1 + a)$. Both lie in E, and

$$a + a^Q = 1 + U + W \in E, \qquad a^2 + (1 + U + W)a + U = 0.$$

Thus a has degree at most two over E and belongs to $\mathbb{F}_{2^{2s}} \subset L$. The same holds for $1 + a$.

Write $t = 2^s$ and $m = r/s$. Cyclic subgroup intersection gives $C_d \cap \mathbb{F}_{t^2}^{\times} = C_{\gcd(d, t^2-1)}$. If m is even, $t^m + 1 \equiv 2 \pmod{t^2 - 1}$, so this gcd is u. If m is odd, $t^m + 1 \equiv t + 1 \pmod{t^2 - 1}$; since $u \mid t - 1$ and $\gcd(t - 1, t + 1) = 1$, the gcd is $u(t + 1)$. This proves one inclusion in (6.9); the other follows from $C_g \subset C_d$. In both cases $g < d$, so every pair in the left-hand set generates a proper subgroup of C_d. Therefore $\mathcal{R}_d = \varnothing$.

The group C_d contains C_{2^r+1}, whose action on L is irreducible by the field-degree proof in Theorem 6.5. Hence H_d is primitive. Corollary 2.5 excludes scalar realizations, and every affine permutation in the specified group is scalar. Linear-conjugacy equivariance of F excludes realizations of its linear conjugates as well. □

Corollary 6.7 (The quarter-power barrier)**.** *For every $s \ge 1$, put $t = 2^s$, $q = t^4$, and*

$$k = t + 1 = q^{1/4} + 1, \qquad d = (t - 1)(t^2 + 1) = \frac{q - 1}{k}.$$

Then H_d is a proper primitive nonrealizable affine companion group, and its membership pairs are exactly

$$\{a \in C_d : 1 + a \in C_d\} = \mathbb{F}_t \setminus \{0, 1\}. \qquad T_d = t - 2, \quad |\mathcal{R}_d| = 0. \tag{6.10}$$

Consequently no exponent $\beta > 1/4$ can give universal eventual realizability for all admissible indices $1 < k \le q^{\beta}$. Even the range $1 < k \le Cq^{1/4}$ fails for every fixed $C > 1$.

Proof. Apply Theorem 6.6 with $r = 2s$ and $u = t - 1$. Here $g = t - 1$ and $C_g = \mathbb{F}_t^{\times}$, yielding (6.10). The indices are proper and $k/q^{1/4} = 1 + 1/t \to 1$. Hence these infinitely many exceptions eventually lie in either range in the statement. □

Together, Corollaries 6.4 and 6.7 determine the exponent $1/4$ for a universal growing-index guarantee. They leave a factor of $\sqrt{n} = \sqrt{\log_2 q}$ between the sufficient range and the obstructed family. In particular, the exact boundary $k \le q^{1/4}$ has not been decided by these arguments.

7 Scope and further questions

The norm-fiber method separates three scales: the $O(t)$ population of one fiber, the $O(t^2)$ population of the full intersection, and the $O(t)$ trace-one subfamily relevant to the relative-trace classification. The populations below make this distinction concrete. Here T_d counts subgroup membership, N_d^{prim} requires only a to have order d, and Δ_t is the Frobenius-coupled subfamily.

s	T_d	$\lvert\mathcal{R}_d\rvert$	N_d^{prim}	D_d	Δ_t
2	24	24	16	8	8
3	86	72	48	24	0
4	240	240	224	208	16
5	1214	960	520	240	0
6	4104	3960	2928	1992	24

These exact finite counts illustrate the three scales distinguished by the density estimates and the relative-trace classification. The single full-order estimate, the four-torsion symmetry of good fibers and its separate existence proof are retained in Supplementary Section S6.

The exceptional groups have different arithmetic causes. At $s = 3$, reciprocal traces of orders $1, 3, 9$ cannot support the required order 65. At $s = 5$, the quotient by C_{11} forces reciprocal-trace order 3 or 33, and the surviving order-33 parameters lose a factor 5. The inverse-support reduction expresses this distinction as an empty primitive support at $t = 8$ and a support confined to a prime-coordinate kernel at $t = 32$. In the other finite cases, representative positivity and a separating witness rule out the two-prime obstruction. The one- and two-prime hypotheses of that reduction delimit its use in the proof.

The refinement also gives information beyond existence. Proposition 5.5 determines the mean on every fixed-norm class of traces and bounds the variance without the $W(A)$ factor in the pointwise sieve. At $t = 32$, the support $T_1 = C_{11} \setminus \{1\}$ explains both the empty subgroup and the two positive populations, 20 and 18. The zero set is a subgroup, but the count is not constant on its other cosets. For arbitrary nonzero K-norm, the exceptional sets become the scalar cosets in Corollary 5.7, by scaling the norm-one result.

Theorem 4.5 completes the individual inner-norm support problem: no primitive norm fiber is exceptional, even when B is composite. Nonprimitive first norms cannot occur in the double full-order population. This support classification differs from relative-trace slices, whose exceptional groups C_9 and C_{11} remain present. It also leaves the second inner norm unspecified and does not improve the density error. The proof closes the existing mixed-character estimate with the classical prime sieve and fully covered finite witnesses; it does not require the trace classification to prove full-family existence.

The new inner-norm distribution results separate a geometric population from the subsequent order filters. The unsieved variance has main term t, and the cubic second-norm coordinate has a mixed covariance of order $O(W(A)\sqrt{t})$. Its genus-two model and the exclusion of Lang-twisted common components explain this cancellation. The remaining A characters and the two B filters still require mixed estimates. In particular, a full-order variance main term and removal of its $W(A)^2$ loss remain unresolved. The odd-parameter constant improvement in (4.11) does not settle these questions.

Two other questions remain open in the present treatment. First, the sufficient uniform range and the obstructed family leave a factor $\sqrt{\log_2 q}$; the exact universal boundary for admissible $k \le q^{1/4}$ is unresolved here. Second, the argument uses the genus-one geometry of a quartic tower. Extending the norm-dependent coefficient calculation to higher towers requires a corresponding

analysis of singular fibers and the geometric character exceptions.

The companion-group reduction applies to all affine complete mappings. Its linear complement is abelian, so the resulting groups are metabelian. The supplement records the genuinely different nonlinear behavior: an explicit companion pair generates A_{16} (Proposition S4.2). Its coordinate examples and oriented-spread classifications provide further applications of F, with separate statements of computational coverage.

A Certificates for complete primitive inner-norm support

This appendix closes the sieve in Lemma 4.6 for every $s \geq 2$. Throughout the positivity argument $b \in U$ is primitive. The files named below are in `verification/primitive_norm_2026-11-02/`. The proof uses the existing Hasse and mixed-character bounds; the finite records verify the arithmetic inputs to that proof.

A uniform infinite tail

There are 563 odd primes below 4096, and 276 primes below 4096 congruent to one modulo four. If their sets are $\mathcal{P}$ and $\mathcal{P}_1$, respectively, exact integer products give

$$4096^{563} < 2^{12\cdot 79} \prod_{p\in\mathcal{P}} p, \qquad 4096^{276} < 2^{12\cdot 39} \prod_{p\in\mathcal{P}_1} p.$$

For odd n, the factors $4096/p^e$ for $p \geq 4096$ are at most one. Adding missing smaller primes increases the product, so

$$\frac{W(n)^{12}}{n} = \prod_{p^e \| n} \frac{4096}{p^e} \leq \prod_{p\in\mathcal{P}} \frac{4096}{p} < 2^{12\cdot 79}.$$

Every prime divisor of $B = 2^{2s}+1$ is one modulo four, since 2^s has order four modulo that prime. Consequently

$$W(A) < 2^{79} A^{1/12}, \qquad W(B) < 2^{39} B^{1/12}.$$

Using $AB^2 < 8t^5$, we obtain

$$8W(A)W(B)^2 < 2^{160+1/4} t^{5/12} < \sqrt{t} \qquad (s \geq 1924).$$

The last inequality is equivalent to $s > 1923$. With all prime positions in the core, (4.8) is therefore positive throughout this infinite range.

Exact integer sieve certificates

For each $2 \leq s \leq 1923$, divide A and B by all primes at most 4096, with full multiplicities. Let P_A, P_B be the sets of distinct small factors and R_A, R_B the residuals. Choose the uniquely determined nonnegative integers k_A, k_B with

$$4097^{k_A} \leq R_A < 4097^{k_A+1}, \qquad 4097^{k_B} \leq R_B < 4097^{k_B+1}.$$

Each residual prime exceeds 4096, so k_A, k_B bound the numbers of distinct residual primes. No residual is assumed prime.

For core subsets $C_A \subseteq P_A$, $C_B \subseteq P_B$, use C_B in both B coordinates and define

$$\delta_0 = 1 - \sum_{p \in P_A \setminus C_A} \frac{1}{p} - 2 \sum_{p \in P_B \setminus C_B} \frac{1}{p} - \frac{k_A + 2k_B}{4097},$$
$$N_0 = |P_A \setminus C_A| + 2|P_B \setminus C_B| + k_A + 2k_B,$$
$$W_0 = 2^{|C_A|+2|C_B|}, \qquad C_0 = 8W_0 \left(2 + \frac{N_0 - 1}{\delta_0}\right) - 6.$$

When $\delta_0 > 0$, the actual omitted positions satisfy $\delta \geq \delta_0$ and $N \leq N_0$. For $N \geq 1$, monotonicity gives $8W_0\Lambda - 6 \leq C_0$. If $N = 0$, its coefficient $8W_0 - 6$ is also at most C_0: this is equality if $N_0 = 0$, when $\delta_0 = 1$, and follows from $C_0 \geq 16W_0 - 6$ otherwise. In particular $C_0 > 0$. Thus the exact comparison

$$(2^s + 1)^2 \operatorname{den}(C_0)^2 > 2^s \operatorname{num}(C_0)^2 \tag{A.1}$$

proves positivity for every primitive b at that parameter.

The file `global_sieve_controls.json` records all 1922 rows, including factorizations by the small primes, residuals, cores, rational δ_0, C_0 and the integer comparison. There are 1909 successful rows. The unsuccessful sufficient tests are precisely $s \in \mathcal{E}$ from the theorem proof; they do not assert that any fiber is empty. In particular $s = 14$ passes, with

$$\delta_0 = \frac{42864266}{67129345}, \qquad C_0 = \frac{1825425610}{21432133}.$$

The finite search only selects successful cores; no optimality of those cores is used.

Witnesses for every remaining norm

The file `remaining_fiber_witnesses.json` gives one witness per orbit of squaring on the primitive elements of U for each $s \in \mathcal{E}$. The coverage is

s	$\varphi(A)$	orbits	s	$\varphi(A)$	orbits
2	4	1	9	324	18
3	6	1	10	800	40
4	16	2	11	1364	62
5	20	2	12	3840	160
6	48	4	13	5460	210
7	84	6	15	19800	660
8	256	16			

Thus 1182 witnesses cover 32022 primitive norms. A row specifies an irreducible binary polynomial M of degree $2s$, an element u of order A in $E = \mathbb{F}_2[X]/(M)$, and exponents j, k with $\gcd(j, A) = \gcd(k, A) = 1$. Put

$$b = u^j, \quad w = u^k, \quad c = 1 + b + w.$$

The tests verify $c \neq 0$ and $\operatorname{Tr}_{E/\mathbb{F}_2}(b/c^2) = 1$. Hence $L = E[Z]/(Z^2 + cZ + b)$ is a field. For $a = Z$, they check

$$a^B = b, \quad (1 + a)^B = w, \quad a^d = (1 + a)^d = 1,$$
$$a^{d/p} \neq 1, \quad (1 + a)^{d/p} \neq 1 \qquad (p \mid d \text{ prime}).$$

All factors of A, B in these 13 cases are verified by deterministic trial division. The supplied orbit lists are disjoint and their union is exactly the unit exponents modulo A. Frobenius transports a

witness a to a^2, its first norm to b^2, and preserves both orders. These checks therefore cover every primitive norm, rather than a sample of norms. Finite-field isomorphisms transfer the statements from the displayed models to E and L in the theorem, respecting their unique subfields and the relative norms.

Independent replay and proof scope

The verifier `verify_closure_independent.py` imports no producer arithmetic. It reconstructs the primes and integer inequalities, uses Rabin irreducibility and literal quadratic tower powers, and checks 16026 prime-order tests on the two linked elements. Its report is `independent_closure_controls.json`.

A separately written review implementation, `replay_independent.py`, instead uses Berlekamp irreducibility and the normalized Artin–Schreier tower $T^2 + T + b/c^2$, representing $a = cT$. It also checks the other root, transports the witnesses through every orbit, and verifies 96066 transported norm identities on all 32022 norm positions. Its report is `independent_replay_results.json`. The integer-only check `check_sieve_integer.py` independently reconstructs every selected sieve row and both prime products. All three checks replay the supplied data; none calls a witness finder or assumes a residual is prime. The two JSON data files retain their research-archive hashes.

The general character bounds, the prime sieve, the tail inequality and the Frobenius propagation are proved in the paper. These computations certify their finite arithmetic inputs. The added theorem is outside the retained Lean scope.

B Finite certificates and the all-parameter proof

This appendix records finite inputs for the trace classification and a retained alternative proof of full-family existence. The two double-order certificates below complete that alternative route after Theorem 5.4. The stronger complete inner-norm support theorem has its own certificates in Appendix A. The integer tail bound here supplies the range $s \geq 33$ in the trace classification. The complete field-certificate tables for $2 \leq s \leq 32$, including the distinct single-order and double-order records, are retained in Supplementary Appendix S.B.

Two off-trace-one double-order certificates

A hexadecimal integer encodes a binary polynomial with its least significant bit the constant coefficient. In each row set

$$E = \mathbb{F}_2[X]/(M_s), \qquad c = 1 + b + w, \qquad L = E[Z]/(Z^2 + cZ + b), \qquad a = Z.$$

The entries for M_s, b, w, c are hexadecimal; d is decimal. Both rows use $w = b^4$.

s	M_s	b	$w = b^4$	c	d
3	43	6	1c	1b	585
5	409	9b	3e5	37f	33825

The checks certify that M_s is irreducible of degree $2s$ and $\mathrm{Tr}_{E/\mathbb{F}_2}(b/c^2) = 1$, so the quadratic defines a field. In that field they verify

$$a^d = (1+a)^d = 1, \qquad a^B = b, \qquad (1+a)^B = w,$$
$$a^{d/p} \neq 1, \qquad (1+a)^{d/p} \neq 1 \quad (p \mid d \text{ prime}).$$

The prime divisors are $3, 5, 13$ for $s = 3$ and $3, 5, 11, 41$ for $s = 5$; their primality is checked by the recursive Lucas certificates supplied with the data. Since $c \neq 1$, both examples lie outside the trace-one slice. The stored rows are in `verification/author_runtime/inner_double_order_certificates.json`. The endpoint driver `verification/narrative_revision_2026-10-29/verify_two_exceptional_certificates.py` calls the existing deterministic field and order checker on these two rows. The full 31-row tables and their verification details remain in Supplementary Appendix S.B.

An independent integer tail bound

The following arithmetic statement is independent of the existence classification and is used in Theorem 5.4 and Supplementary Proposition S7.1.

Lemma B.1 (Integer tail bound)**.** *For $s \geq 33$, put $t = 2^s$ and $d = (t+1)(t^2+1)$. Then $\sqrt{t} \geq 4W(d)$.*

Proof. We cover $33 \leq s \leq 113$ by exact divisor upper bounds, and $s \geq 114$ by a uniform inequality for odd integers. No large residual factor is assumed to be prime.

For the uniform bound let P be the product of all 53 odd primes less than 256, namely

$$3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83,$$
$$89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173,$$
$$179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251.$$

The exact integer comparison $256^{53} < 4096^8 P$ gives, for every odd n,

$$\frac{W(n)^8}{n} = \prod_{p^e \| n} \frac{256}{p^e} \leq \prod_{3 \leq p < 256} \frac{256}{p} = \frac{256^{53}}{P} < 4096^8.$$

Here the product on the right is over primes: each factor for a prime at least 256 is at most one, and adding missing smaller prime factors only increases the bound. Thus $W(n) < 4096 n^{1/8}$. For $s \geq 114$, $d < 2t^3$ and $t > 2^{113}$ give

$$4W(d) < 2^{113/8} t^{3/8} < \sqrt{t}.$$

For each $33 \leq s \leq 113$, divide A and B by every prime at most 4096, with its full multiplicity. Let j_A, j_B be the numbers of distinct small factors and R_A, R_B the residuals. All prime factors of each residual exceed 4096. If $4097^{k_A} \leq R_A < 4097^{k_A+1}$ and similarly for B, then

$$\omega(d) = \omega(A) + \omega(B) \leq r_s := j_A + k_A + j_B + k_B.$$

The following table lists all 81 bounds; each satisfies $s \geq 4 + 2r_s$, so $\sqrt{t} \geq 4\, 2^{r_s} \geq 4W(d)$.

s	r_s	s	r_s	s	r_s
33	11	60	15	87	24
34	9	61	16	88	21
35	13	62	15	89	23
36	11	63	21	90	27
37	11	64	15	91	26
38	11	65	20	92	23
39	13	66	19	93	25
40	9	67	18	94	24
41	11	68	16	95	26
42	14	69	19	96	24
43	12	70	20	97	26
44	11	71	18	98	26
45	17	72	19	99	29
46	13	73	20	100	26
47	13	74	19	101	26
48	12	75	23	102	27
49	15	76	19	103	26
50	14	77	23	104	25
51	15	78	22	105	32
52	12	79	21	106	27
53	14	80	19	107	28
54	16	81	23	108	29
55	16	82	21	109	27
56	13	83	22	110	29
57	16	84	23	111	30
58	15	85	24	112	27
59	15	86	22	113	29

The full small factorizations, residual integers and their integer power bounds are supplied in `inner_double_omega_certificates.json`. The independent verifier `verify_inner_double_all.py` reconstructs every small prime by deterministic trial division, checks the exact products and all 81 inequalities, and checks $256^{53} < 4096^8 P$. □

The exact products, residual bounds and recursive prime certificates remain in `verification/author_runtime/`. The reproduction guide identifies the corresponding verifiers; Supplementary Appendix S.D retains the full computational record.

C Certificates for all relative traces and the exceptional groups

For Lemma 5.3, the following table lists a squarefree core e and an integer upper bound $C_s \geq 4W(e)\Lambda - 2$. In every row $\delta > 0$ and $(2^s+1)^2 > 2^s C_s^2$, so positivity holds for every $c \in E^*$. The omitted primes are all prime divisors of $(2^s+1)(2^{2s}+1)$ outside e. Full factorizations, the exact rational δ, Λ and both integers in each comparison are recorded in `verification/uniform_trace/uniform_trace_checks.json`.

s	e	C_s	s	e	C_s
8	1	11	21	3	100
10	1	23	22	1	34
11	1	41	23	1	59
12	1	24	24	1	27
13	1	52	25	3	98
14	1	29	26	1	40
15	3	96	27	3	103
16	1	15	28	1	24
17	1	50	29	1	51
18	1	52	30	1	58
19	1	50	31	1	58
20	1	19	32	1	19

No asymptotic extrapolation from this table is used: $s \geq 33$ is supplied by the independent integer bound of Lemma B.1 and Theorem 5.1.

For the finite trace-one counts, define H_A as in (5.12). In the next table an entry $e : j$ means $\deg \gcd(H_A, X^{B/e} + 1) = j$; only nonzero degrees are listed. Every unlisted squarefree divisor of B has degree zero. Möbius inversion gives the last column.

s	$\deg H_A$	Nonzero gcd degrees $e : j$	Δ_t
2	8	$1 : 8$	8
3	12	none	0
4	32	$1 : 16$	16
5	40	$1 : 20,\ 5 : 20$	0
6	96	$1 : 24$	24
7	168	$1 : 84$	84
8	512	$1 : 288$	288
9	648	$1 : 360,\ 5 : 36,\ 13 : 72,\ 37 : 36$	216

For the two exceptions the reciprocal-trace polynomials are encoded by the hexadecimal integers

$$H_9 = \texttt{13f9}, \qquad H_{33} = \texttt{1950fbbe153}.$$

The verifier constructs the cyclotomic polynomials by exact division, computes every modular power and gcd, and checks explicit polynomial identities $uH_A + v\Phi_B = 1$ at $s = 3, 5$. The coefficients u, v are archived as binary-polynomial integers. It additionally checks the degree-20 order drop at $s = 5$ and the coprimalities excluding orders 5 and 41.

The standard-library verifier is `verification/uniform_trace/verify_uniform_traces.py`. Its independent small-field checks enumerate all nonzero K-valued relative traces for $2 \leq s \leq 6$ in two ways: through the full norm-one subgroup and through the reciprocal-trace map on V. They agree on both Q_c and M_c, on the binary Kloosterman identity and on all stated numerical error bounds. The analytic character estimate and the infinite parameter range rely on the proofs, not on these finite checks.

Representative and separating support witnesses

Write $\rho = \mathrm{rad}(A)$ and choose generators $z \in E^*$, $\Gamma \in V$. Represent $h = z^{Aj}$, $0 \leq j < t - 1$. Each certificate records k and e with

$$\gcd(k, B) = 1, \qquad \Gamma^k + \Gamma^{-k} = h^2 (z^{t-1})^e.$$

A central witness has $\gcd(e, A) = 1$ and proves $M_h > 0$. When A has two distinct prime divisors, a separating witness has exponent e' with $e' \not\equiv e, -e \pmod{\rho}$. Inversion supplies the second residue $-e$, so these two witnesses certify at least three distinct points in the radical quotient. The inverse-support lemma then proves positivity on the whole trace class hU. Only these witnesses, not the complete support populations, are required by this step of the proof.

s	A	ρ	Central witnesses	Separating witnesses
2	5	5	3	0
3	9	3	6	0
4	17	17	15	0
5	33	33	30	30
6	65	65	63	63
7	129	129	127	127
9	513	57	511	511
Total			755	731

The omitted representatives are precisely $(s, h) = (3, 1), (5, 1)$, whose supports are determined by the polynomial identities. The case $s = 8$ is already analytic; 255 additional central witnesses are retained as a finite check. Thus the archive contains $1010 + 731 = 1741$ witnesses, of which 1486 are used by the finite part of the current proof.

The exact data are in `verification/radical_support_2026-10-26/support_certificates.json`. The producer enumerates reciprocal traces from primitive γ and records central and separating exponents. The independent verifier uses single binary polynomial fields of degree $4s$, checks irreducibility and the field embeddings, and directly verifies every reciprocal trace and order condition. It constructs a full-order element for each central witness. The two exceptional supports are separately enumerated at $s = 3, 5$. Neither verification nor the group-theoretic proof uses the truth of the completed trace classification.

The older all-E certificate family below is retained for cross-checking. It already compressed traces into witness families; the improvement here is the reduction to representative and separating support witnesses, without checking every trace residue for coverage. Excluding the analytic $s = 8$ case, the older proof used 2809 family records, compared with 1486 support witnesses in the present finite argument.

Retained witness families on the whole trace field

Put $r = t - 1$ and $N = t^2 - 1 = rA$, and choose a primitive $z \in E^*$ and a generator $\Gamma \in V$. Write every $c \in E^*$ uniquely as $c = z^{\ell + rm}$, with $0 \le \ell < r$ and $0 \le m < A$. For a recorded exponent k with $\gcd(k, B) = 1$, certify

$$\Gamma^k + \Gamma^{-k} = z^j, \qquad j \equiv 2\ell \pmod{r}.$$

Then $w = z^{j - 2\ell - 2rm} \in U$, and it has order A exactly when

$$\gcd\left(\frac{j - 2\ell}{r} - 2m, A\right) = 1.$$

For each such residue m, Theorem 5.1 constructs a full-order element $a = c/(1 + \Gamma^k)$. The recorded lists of k cover every residue outside the two exceptional groups, as follows. "Families" counts recorded values of k; it is not a count of individual trace slices.

s	$\lvert E^* \rvert$	Nonempty traces	Families	Empty traces
2	15	15	6	none
3	63	54	12	C_9
4	255	255	30	none
5	1023	1012	117	C_{11}
6	4095	4095	227	none
7	16383	16383	484	none
8	65535	65535	510	none
9	262143	262143	1933	none

The coefficients, primitive generators, logarithms j and lists are in `verification/extended_trace_2026-10-26/extended_trace_certificates.json`. The generator uses the exact recurrence $\tau_{k+1} = \tau_1\tau_k + \tau_{k-1}$ for $\tau_k = \Gamma^k + \Gamma^{-k}$. The independent verifier instead works in single binary polynomial fields of degree $4s$. It checks their irreducibility and the field isomorphisms, verifies every displayed logarithm by direct powering, and tests every coverage residue by the gcd condition above. For each of the 3319 families it also constructs an element and directly verifies its relative trace, norm and exact order. Thus 349492 nonempty traces are certified; the remaining $9 + 11$ traces are proved empty in Theorem 5.4.

For $s = 2, \ldots, 6$, independent enumeration of the entire subgroup checks all 5451 nonzero E-valued populations, both stated error bounds and $Q_c = Q_h$ for $h^2 = N_{E/K}(c)$. It also confirms that M_c need not equal M_h. The verifier and exact output are `verification/extended_trace_2026-10-26/verify_extended.py` and `independent_checks.json` in that directory.

The earlier 1010 individual witnesses for the K^* restriction remain archived in `verification/uniform_trace/`, with their separate binary-field audit in `verification/proof_audit_51_54_2026-10-26/`. The trace-one polynomial counts and integer tail certificates are retained. Positivity in the infinite range follows from the analytic proof, not from extrapolating these finite checks.

D Data, verification and formalization scope

The accompanying source package contains the exact certificates, deterministic verifiers, field representations and element-level data logs. The preceding public baseline is archived at `https://doi.org/10.5281/zenodo.23757428`. The present revision adds complete primitive inner-norm support and the primitive-norm distribution results of Section 4.5; its new sources, certificates and replay instructions are supplied in this package. Historical guides and outputs retain their dates and describe their corresponding earlier versions. The repository and pinned original input revision are specified in Supplementary Appendix S.D.

Inputs to the complete support and trace proofs

Result or range	Exact input	Location and verification
Theorem 4.5, $s \geq 1924$	Two exact small-prime products and the full-core inequality	Appendix A; both integer implementations reconstruct the products.
$2 \leq s \leq 1923$	1922 integer rows, of which 1909 prove positivity	`global_sieve_controls.json`; exact rational comparisons and residual prime budgets, with no unproved primality assumption.
$s \in \mathcal{E}$	1182 witnesses covering all 32022 primitive inner norms	`remaining_fiber_witnesses.json`; independent quadratic tower implementations and complete Frobenius-orbit coverage.
Theorem 5.4, $s = 2, \ldots, 7, 9$	755 central and 731 separating witnesses; exceptional polynomial identities	Appendix C; raw data in `verification/radical_support_2026-10-26/`.
$s = 8, 10, \ldots, 32$	24 prime-sieve rows	Appendix C; the tower-field closure checker verifies their exact rational inequalities.
$s \geq 33$	81 integer rows for $33 \leq s \leq 113$ and the analytic bound for $s \geq 114$	Lemma B.1; `verify_inner_double_all.py` checks the finite integer inputs.
Alternative full-family existence, $s = 3, 5$	The two displayed off-trace-one double-order certificates	Appendix B; `verify_two_exceptional_certificates.py` checks both rows.

The complete primitive inner-norm theorem uses the mixed-character estimate and the three-coordinate sieve, explicitly with primitive b. Its three arithmetic ranges exhaust all $s \geq 2$ and prove full-family existence independently of the trace classification. The separate trace proof proceeds from the uniform relative-trace estimate through its prime sieve, support certificates and integer tail bound to the complete exceptional groups. The two off-trace-one certificates complete that alternative existence route. The character estimates and infinite-range arguments are proved in the paper. The full field tables, additional enumerations and retained alternative proofs are documented in Supplementary Appendices S.B and S.D. `REPRODUCE.txt` gives the exact project-relative commands; `delivery_manifest.json` records the delivered file hashes.

Distribution checks. Supplementary Appendix S.E proves the Kloosterman identity, the genus-two reduction and the mixed moment. The accompanying independent arithmetic implementations check the cubic reduction at all 1798 primitive norms for $s = 3, 5, 7, 9, 11$. An exact spectral verifier reconstructs the two Frobenius polynomials and all 54 cyclotomic gcds used in the projective specialization argument. These finite checks support the explicit inputs; the geometric implication and the infinite-parameter estimates are proved in the text. Fresh replay reports are in `verification/general_revision_2026-11-02/`.

Selective formalization. The retained Lean 4.31.0 record covers nine modules with 129 theorem or lemma declarations and 33 definitions. It includes the rational a–γ equivalence, exact-order components and the transfer to the original M_c population through named `Algebra.norm` and `Algebra.trace`. Concrete field certificates, the geometric character estimates, the complete exceptional-zero classification, complete primitive inner-norm support the new distribution and mixed-moment arguments, and the all-parameter assemblies remain outside Lean. Supplementary Appendix S.D and `formalization/INTERFACE_SCOPE.txt` give the pinned environment, axiom audits and exact scope.

E Comparison with earlier prescribed-order results

We compare the relative-trace theorems with the closest prescribed-order, trace and primitive-normal frameworks. Throughout, $R = t-1$, $A = t+1$, $B = t^2+1$ and $d = AB$. The issue is the effective error and the complete classification of individual traces, since the counting object itself is prior.

E.1 Prescribed order and trace

The counting problem itself has direct predecessors. With $q_0 = t^2$, $n = 2$, $\ell = t-1$, one function $f(X) = X$ and prescribed trace c, it is exactly $N_{\ell,1}$ of [22, §2]; it is also $P_{t^2,2,d}(c)$ of [23, §6]. The zero additive-frequency term in [22, eq. (2.2)] is

$$\frac{\varphi(d)}{t^2} = \theta(d)(t+1) + \theta(d)(t^{-1}+t^{-2}),$$

so the leading linear main term is already present in that framework. However, the bracket in its Theorem 2.4 specializes to $1-2(t-1)W(d) < 0$. That bound gives no positivity here; (5.5) obtains the smaller error scale $W(d)\sqrt{t}$ by using the curve over K.

One can also lift $a = \beta^{t-1}$ with β primitive in L, apply the primitive sieve to $f(X) = X^{t-1}$, and divide the count by $\varphi(t-1)$. In the notation of [22, Theorem 3.1], this gives $D = (q_0-1)(t-1)$, $E = q_0 - 1$ and $\delta_n = 0$. Its sufficient inequality would imply

$$q_0 > (q_0-1)(t-2) + q_0^{-1} > q_0 \qquad (t \geq 4),$$

since its squarefree-core sieve factor is at least one. Thus neither of these direct specializations supplies the present conclusion. This comparison concerns the stated bounds, rather than every refinement of their methods.

The earlier function-field framework of [10] includes the same count. In its Theorem 4.3 take $q = Q = t^2$, $k = 1$, $m = 2$, $r = 1$, $f_1(X) = X$ and index $a = R = t-1$. Strong linear independence holds because each nonzero linear combination has a simple pole. Write u, v for its two coprime squarefree sieve factors, so $uv = \mathrm{rad}(d)$, and put $z = uR \geq 3$. Each linear Artin–Schreier subcover has genus zero. Its displayed sufficient condition therefore requires

$$\theta(v) > 1 - \frac{1}{Q} + \frac{2(Q-1)(z-1)Q + Q(2+z) - 2}{Q(Q^2-1)} > 1,$$

which is impossible for every allowed core.

The rough genus bound is not essential to this failure. Following [10, Remark 4.4 and proof of Theorem 4.3], retain the exact genus of $y^Q - y = \alpha^{vR}x^{uR} + \delta$. Since z is odd, it is $(Q-1)(z-1)/2$. The improved threshold becomes

$$1 - \frac{1}{Q} + \frac{z-1}{Q+1} + \frac{Q(2+z)-2}{Q(Q^2-1)} > 1.$$

Indeed $z \geq 3$ and $2/(Q+1) > 1/Q$. This also rules out the primitive lift: use index 1 and $f_1(X) = X^R$, and divide its count by $\varphi(R)$. Its exact composed-cover genus is the same. Encoding the E-valued trace by two K-valued traces also gives no improvement: take $q = t$, $k = m = 2$, $r = 2$ and functions $X, \xi X$, with $\xi \in E \setminus K$. The trace pairing makes these two coordinates equivalent to the prescribed relative trace. The cover still has degree $q^r = Q$ and the same exact genus, while the pole-degree correction increases. Thus neither the choice of sieve core nor this change of trace coordinates repairs the criterion.

The asymptotic statements [10, Theorems 4.5–4.6] do not supply the missing uniform conclusion. Theorem 4.6 does allow a growing order index, but fixes the coefficient field and the trace functions, including their number. Its large-extension assertion does not specify a relative trace in a growing quadratic subfield. Re-encoding that value by binary traces requires a growing number of functions, outside those fixed-data quantifiers.

There is also a superficial coincidence of exceptional exponents. In [10, Example 5.1 and Theorem 5.2], the primitive element lies in $\mathbb{F}_{2^m}$ and satisfies two absolute-trace conditions for fixed rational functions; the exceptional exponents include $3, 5$. Taking $m = s$ gives order $t-1$ in K, whereas the present problem has order d in L. Taking $m = 4s$ gives the ambient field L but a different order and different trace constraints. At $s = 2$ that example has no admissible primitive element, whereas $M_1 = 8$ here. The two existence statements are therefore different; no conversion to the relative-trace classification is supplied by those results.

The line framework of [6, criterion (9)] also counts M_c: use base E, extension degree two, index $t-1$, and the affine line of relative trace c. Its displayed sufficient condition becomes $t > (t-1)W(d)$, which fails for $t \geq 4$. Thus the counting object and the line interpretation are prior.

The reciprocal-trace support viewpoint also has a direct predecessor. [24, Theorem 4.3 and §4.1] describes the possible orders and traces of iterates of $\gamma + \gamma^{-1}$ for $\gamma \in V \setminus \{1\}$; its Theorem 4.12 records a related unfiltered Kloosterman count. Those results do not impose the primitive input and the c-dependent order condition simultaneously. The additional step here is the effective filtered estimate and the inverse-support analysis of its complete zero set, rather than the reciprocal-trace substitution or the general support perspective.

E.2 Product, hypersurface and rational-image criteria

The product criterion in [25, Theorem 1.5(1)] fails for every allowed decomposition of the index, not just a selected one. Use base E and extension degree two. Put $R = t-1$, $m = t^4 - 1$, and $\vartheta = \varphi(m)/m$. For $R = r_1 + \cdots + r_k$ with all $r_i \mid m$, their notation gives $\Lambda_{r_i} = \varphi(m)/\varphi(m/r_i)$ and $\Lambda_R = \varphi(R)$. Every r_i is odd, so $k \geq 3$ is odd. Each $a_i = \vartheta/\Lambda_{r_i}$ satisfies $a_i \leq \vartheta \leq 2/3$ and $a_i \leq 1/r_i$. One r_i is at least R/k, and hence the squared ratio of the two sides of their required inequality is at most

$$\frac{\vartheta^{k+1} t^2}{2\Lambda_R \prod_i \Lambda_{r_i}} = \frac{\theta(d) t^2}{2R} \prod_i a_i \leq \frac{\theta(d) t^2}{2R^2} k (2/3)^{k-1} \leq \frac{2\theta(d)}{3} \left(\frac{t}{t-1} \right)^2 < 1.$$

The last inequality holds for $t \geq 8$; for $t = 4$, use $\theta(d) = 64/85$. Their sign-disjointness refinement is unavailable in characteristic two, since $-y = y$. Their complete 3-primitive classification [25, Theorem 1.6] does imply positivity in the present $s = 2$ case, by taking base $E = \mathbb{F}_{16}$ and degree two. It does not classify the growing index $t-1$.

The primitive-point framework of [26] is also relevant. Its Fermat-hypersurface count, after primitive lifting, includes two prescribed-order elements with prescribed sum. It does not impose the additional conjugacy $b = a^{t^2}$ that defines M_c. For example, the adjacent-pair counts at $t = 8, 32$ are $24, 240$, respectively, while $M_1 = 0$ in both fields. Its Dwork-regular estimate requires degree prime to the characteristic; the natural norm polynomial here has degree four in characteristic two. Its exact hyperplane formula assumes an odd Fermat-prime field. These results do not directly give the four-point bound or the exceptional relative-trace groups.

The rational-image framework of [27] contains these counts as well, including its prescribed inverse trace. Over base E, put $q = t^2$, $m = 2$ and

$$F_c(X) = \frac{X}{X + c}, \qquad r_1 = t - 1, \quad r_2 = t^2 - 1, \quad f_1 = f_2 = 1.$$

Both numerator and denominator are irreducible, coprime linear polynomials, and the denominator is not divisible by X. On the trace-c slice, $F_c(a) = a^{1-t^2}$. If a has order d, then $F_c(a)$ has order B, and both are normal over E. Let $\mathcal{C}_c(b)$ be their count $C_{F_c,c,b}(d, B, 1, (X-1)^2)$ with these parameters. The two auxiliary elements are uniquely determined because $f_1 = f_2 = 1$. Since $\mathrm{Tr}_{L/E}(a^{-1}) = c\,a^{-B}$, we have exactly

$$M_c = \sum_{\substack{w \in U \\ \mathrm{ord}(w)=A}} \mathcal{C}_c(cw).$$

Here $X^2 - 1 = (X-1)^2$, so their factor coprime to $X-1$ is 1 and the remaining additive squarefree-divisor factor is 2. Their Corollary 3.2, equation (3.9), would require

$$t^{-2} > 10(t-1)^2(t+1)W(d)W(B),$$

which is impossible. Their Theorem 3.3 retains the same left side for every sieve core, while its right side is at least $5(t-1)(t^2-1)$: its sieve factor $\Delta \geq 1$. More generally, all their displayed general and sieve criteria have left side $q^{m/2-k_1-k_2-2} \leq 1$ when $m \leq 4$, and right side at least $n_1 + 3n_2 + 1 \geq 2$. Thus using base K and $m = 4$ cannot repair the bound; it also prescribes only the coarser trace $c + c^t$. Their Lemma 3.5 and Theorem 4.3 require $m > 2(k_1 + k_2 + 2)$, and their numerical section treats $m \geq 7$. These comparisons retain the prior counting framework without attributing the present effective estimate or exceptional groups to its stated results.

The prescribed-norm framework of [21] also contains these counts. Put $m_{b,c} = \#\{a \in L : \mathrm{ord}(a) = d, N_{L/E}(a) = b, \mathrm{Tr}_{L/E}(a) = c\}$. Then $M_c = \sum_{\mathrm{ord}(b)=A} m_{b,c}$, and each $m_{b,c}$ is a specialization of their Section 4 count: for a fixed primitive lift $\beta \in E$ with $\beta^{t-1} = b$, the power map has a unique preimage with norm β, since $\lambda \mapsto \lambda^2$ permutes the $(t-1)$st roots of unity. Their effective conditions do not yield positivity here. For the K-valued slices, take base K, $n = 4$, $r = t-1$, $k = 1$, $g = X - 1$, norm 1 and trace 0; over E, for any $c \in E^*$, take $n = 2$, $r = t-1$, $k = 0$, $g = 1$, norm b of order A and trace $c \neq 0$. The repeated-factor convention in [21, Theorem 4.1 and Corollary 4.1] gives, respectively,

$$t^{-1} > (t-1)W(d), \qquad t^{-2} > (t-1)W(B),$$

both impossible for $t \geq 4$. The first count also aggregates the nonzero K-valued relative traces. Their Proposition 4.1 fixes r, k and assumes $n \geq 4k + 5$; neither specialization lies in that range. Their Section 4.1 gives large-field sufficient bounds, rather than the exceptional slices classified in Theorem 5.4.

E.3 Primitive-normal aggregates and projective geometry

Hachenberger's earlier quartic theorem is a geometric predecessor [11, Theorem 1 and §3]. For even t it gives

$$PN_4(t) \geq \varphi(t^4 - 1) - \varphi(t-1)\varphi(t^2+1).$$

Its projective spread counts primitive points on primitive lines, excluding their trace-zero hyperplane intersections. In our notation the exact bridge to that aggregate is

$$PN_4(t) = \varphi(t-1) \sum_{c \in E \setminus K} M_c.$$

Indeed $L^* = K^* \times C_d$, and every generator of C_d has exactly $\varphi(t-1)$ primitive ambient-field lifts. Scaling by K^* preserves normality. In characteristic two and degree four, normality over K is equivalent to $\mathrm{Tr}_{L/K}(a) \neq 0$, or $c + c^t \neq 0$. Consequently its lower bound is equivalent to

$$\sum_{c \in K^*} M_c \leq \varphi(B).$$

It bounds an aggregate rather than every prescribed trace. For example its Table 1 gives $PN_4(8) = 1512$ and $PN_4(32) = 465000$, while $M_1 = 0$ at both parameters. These aggregate counts are consistent with the two empty trace-one slices and do not classify the exceptional groups. The later norm-one and ovoid model [12] provides further geometric context; Supplementary Appendix S.C compares its trace sections with the additional norm-to-trace ratio fixed here.

The comparisons in this appendix concern stated results and the explicit specializations above. They credit the established counting frameworks, geometric methods and cyclic-group tools; they do not rule out every possible further refinement of those methods or assert absolute global priority.

References

[1] M. Hall, L. J. Paige, *Complete mappings of finite groups*, Pacific J. Math. 5 (1955) 541–549, doi:10.2140/pjm.1955.5.541.

[2] B. D. McKay, I. M. Wanless, *Enumeration of Latin squares with conjugate symmetry*, J. Combin. Des. 30 (2022) 105–130, doi:10.1002/jcd.21814.

[3] P. van Wamelen, *Jacobi sums over finite fields*, Acta Arith. 102(1) (2002) 1–20, doi:10.4064/aa102-1-1.

[4] N. M. Katz, *An estimate for character sums*, J. Amer. Math. Soc. 2 (1989) 197–200, author's copy.

[5] R. R. Farashahi, P.-A. Fouque, I. E. Shparlinski, M. Tibouchi, J. F. Voloch, *Indifferentiable deterministic hashing to elliptic and hyperelliptic curves*, Math. Comp. 82 (2013) 491–512, doi:10.1090/S0025-5718-2012-02606-8. The Artin-character bound is cited from the author's version, §4, Lemma 1.

[6] S. D. Cohen, G. Kapetanakis, *Finite field extensions with the line or translate property for r-primitive elements*, J. Aust. Math. Soc. 111 (2021) 313–319, doi:10.1017/S1446788720000099.

[7] S. D. Cohen, G. Kapetanakis, L. Reis, *The existence of $\mathbb{F}_q$-primitive points on curves using freeness*, C. R. Math. 360 (2022) 641–652, doi:10.5802/crmath.328.

[8] J. J. R. Aguirre, V. G. L. Neumann, *Pairs of r-primitive and k-normal elements in finite fields*, arXiv:2210.11504v1 (2022), arXiv:2210.11504.

[9] J. J. R. Aguirre, A. Lemos, V. G. L. Neumann, S. Ribas, *r-primitive k-normal elements in arithmetic progressions over finite fields*, arXiv:2211.02114v2 (2023), arXiv:2211.02114.

[10] F. Özbudak, *Elements of prescribed order, prescribed traces and systems of rational functions over finite fields*, Des. Codes Cryptogr. 34 (2005) 35–54, doi:10.1007/s10623-003-4193-0.

[11] D. Hachenberger, *Primitive normal bases for quartic and cubic extensions: a geometric approach*, Des. Codes Cryptogr. 77 (2015) 335–350, doi:10.1007/s10623-015-0051-0.

[12] D. Hachenberger, *Ovoids and primitive normal bases for quartic extensions of Galois fields*, J. Algebr. Comb. 53 (2021) 85–114, doi:10.1007/s10801-019-00920-8.

[13] J. Sharma, R. Sarma, S. Laishram, *Arithmetic progressions of r-primitive elements in a field*, Bull. Braz. Math. Soc. (N.S.) 55 (2024), Article 40, doi:10.1007/s00574-024-00412-9.

[14] A. Choudhary, J. Sharma, *Arithmetic progressions of some special elements in finite fields*, Commun. Algebra 54(4) (2026) 1502–1518, doi:10.1080/00927872.2025.2561951.

[15] H. Kudo, Y. Nogata, *Cyclotomic numbers of order $q-1$ over $\mathbb{F}_{q^r}$*, arXiv:2604.25461 (2026), `https://arxiv.org/abs/2604.25461`.

[16] J. D. Dixon, B. Mortimer, *Permutation Groups*, Springer, 1996.

[17] S. D. Cohen, *The orders of related elements of a finite field*, Ramanujan J. 7 (2003) 169–183, doi:10.1023/A:1026295128600.

[18] A. Bors, Q. Wang, *Cycle types of complete mappings of finite fields*, J. Algebra 591 (2022) 577–610, doi:10.1016/j.jalgebra.2021.09.017.

[19] A. Bors, Q. Wang, *Compositions and parities of complete mappings and of orthomorphisms*, J. Combin. Theory Ser. A 196 (2023) 105723, doi:10.1016/j.jcta.2022.105723.

[20] H. Shin, *Coset-refined trace statistics, nodal characters, and affine branches in cubic norm tori*, arXiv:2605.21939v1 (2026), arXiv:2605.21939.

[21] M. Rani, A. K. Sharma, S. K. Tiwari, A. Panigrahi, *On r-primitive k-normal elements with prescribed norm and trace over finite fields*, Finite Fields Appl. 91 (2023) 102253, doi:10.1016/j.ffa.2023.102253.

[22] S. D. Cohen, *Finite field elements with specified order and traces*, Des. Codes Cryptogr. 36 (2005) 331–340, doi:10.1007/s10623-004-1724-2.

[23] A. Tuxanidy, Q. Wang, *On the number of N-free elements with prescribed trace*, arXiv:1409.6961v2 (2015), arXiv:1409.6961v2.

[24] S. Ugolini, *Some notes on the multiplicative order of $\alpha + \alpha^{-1}$ in finite fields of characteristic two*, Period. Math. Hungar. 80 (2020) 81–94, doi:10.1007/s10998-019-00296-z.

[25] A. Shukla, S. K. Tiwari, *Trace of subset of product of sets and r-primitive elements in finite fields*, Finite Fields Appl. 115 (2026) 102864, doi:10.1016/j.ffa.2026.102864.

[26] J. A. Oliveira, M. Veloso, *On $\mathbb{F}_q$-primitive points on hypersurfaces*, Finite Fields Appl. 118 (2027) 102925, doi:10.1016/j.ffa.2026.102925.

[27] A. Choudhary, J. Sharma, S. K. Tiwari, K. Chatterjee, *Rational image of r-primitive k-normal elements with prescribed traces*, Comput. Appl. Math. 45 (2026), Article 370, doi:10.1007/s40314-026-03761-x.

[28] H. Vemuri, *Domination in direct products of complete graphs*, arXiv:1908.02445v1 (2019).

[29] N. M. Katz, *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, Annals of Mathematics Studies 116, Princeton University Press, 1988, author's copy.

[30] The Stacks Project Authors, *The Stacks Project*, Tags 04DY and 0BQN, topological invariance of the étale fundamental group, accessed 2 November 2026.

[31] B. Li, *An affine Singer layer on $\mathbb{F}_2^6$, its (T, D, K) triple, and $\langle T, D\rangle = \mathrm{AGL}(1, 64)$*, preprint, 2026, doi:10.5281/zenodo.22731492.